(AUTONOMOUS)
NAAC 'A' GRADE

SYLLABUS FOR B.E. CSE

Batch 2023 onwards

Semester 1

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
1	BSC-101	Engineering Mathematics-I	BSC	5	4	1	0	50	100	150
2	BSC-102	Engineering Physics	BSC	5	4	1	0	50	100	150
3	COM-101	Introduction to C Programming	ESC	5	4	1	0	50	100	150
4	HSMC-101	Design Thinking	HSMC	3	2	1	0	50	100	150
5	BSC-112	Engineering Physics Lab	BSC	1	0	0	2	50	0	50
6	COM-111	C Programming Lab	ESC	2	0	0	4	50	0	50
8	NCC-101	Induction Training	NCC	0	0	0	4	-	-	S/NS*
9	MCC-102	Indian Constitution	MCC	1	1	0	0	50	-	50
			Total	22	30					750

*Note: S=Satisfactory, NS=Not Satisfactory

Semester 2

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
1	BSC-201	Engineering Mathematics-II	BSC	5	3	2	0	50	100	150
2	COM-201	Data Structures using C	PCC	4	3	1	0	50	100	150
3	ESC-201	Basic Electrical and Electronics Engineering	ESC	4	3	1	0	50	100	150
4	HSMC-201	Technical Communication	HSMC	4	4	0	0	50	100	150
5	COM-211	Data Structures using C Lab	PCC	2	0	0	4	50	0	50
6	ESC-211	Basic Electrical and Electronics Engineering Lab	ESC	1	0	0	2	50	0	50
8	COM-212	Business Process Automation with RPA Lab	ESC	2	0	0	4	50	0	50
9	NCC-201	Environment and Sustainability	NCC	0	2	0	0	-	-	S/NS*
			Total	22	30					800

*Note: S=Satisfactory, NS=Not Satisfactory.

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Total Marks
1	COM-303	Professional Industrial Training / Internship	Summer Internship	3	0	0	0	100

Note: Students are required to complete the professional industrial training / internship during summer semester breaks.

Semester - 3

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
1	COM-301	Design and Analysis of Algorithms	PCC	5	4	1	0	50	100	150
2	ESC-301	Digital Electronics	ESC	4	3	1	0	50	100	150
3	COM-302	Operating Systems	PCC	4	3	1	0	50	100	150
4	BSC-301	Discrete Mathematics	BSC	3	2	1	0	50	100	150
5	HSMC-301	Principles of Management	HSMC	2	2	0	0	50	100	150
6	ESC-311	Digital Electronics Lab	ESC	2	0	0	4	50	0	50
7	COM-312	Operating Systems Lab	PCC	2	0	0	4	50	0	50
8	HSMC-312	Placement overview and Career Planning	HSMC	1	0	0	2	50	0	50
9	MNC-XXX	MOOC*	NCC	0	0	0	0	-	-	S/NS*
Total				**26***	**28**					**1000***

*Note**:** MOOCs List is to be prepared by the department level committee.
S=Satisfactory, NS=Not Satisfactory
Summer Internship Credits/Marks are also added in this semester's marksheet.

Hons. / Minor Degree Course

Course Code	Specialisation Track	Course Type	Cd	L	T	P	Marks		
							Mid-Semester	End-Semester	Total
MDC-AI/CS/CC/IoT/DS-301	Artificial Intelligence and Machine Learning Cyber Security IoT Data Science Cloud Computing	MOOC	5	5	0	0	25	125	150

*Note: Student can enroll in only one course from chosen Hons. /Minor Degree Course. Total Marks is 150 for any one of the chosen course in the Hons. / Minor Degree Course.

Semester - 4

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
1	COM-401	Object Oriented Programming using C++	PCC	4	3	1	0	50	100	150
2	COM-402	Relational Database Management System	PCC	4	3	1	0	50	100	150
3	COM-403	Computer Organization and Architecture	PCC	5	3	2	0	50	100	150
4	HSMC-401	Entrepreneurship and Business Strategies	HSMC	3	2	1	0	50	100	150
5	COM-411	Object Oriented Programming using C++ Lab	PCC	2	0	0	4	50	0	50
6	COM-412	Relational Database Management System Lab	PCC	2	0	0	4	50	0	50
7	HSMC-412	Corporate Etiquette and Personality Development	HSMC	1	0	0	2	50	0	50
Total				21	26					750

Hons. / Minor Degree Course

Course Code	Specialisation Track	Course Type	Cd	L	T	P	Marks		
							Mid-Semester	End-Semester	Total
MDC-AI/CS/CC/IoT/DS-401	Artificial Intelligence and Machine Learning Cyber Security IoT Data Science Cloud Computing	MOOC	5	5	0	0	25	125	150

*Note: Student can enroll in only one course from chosen Hons. /Minor Degree Course. Total Marks is 150 for any one of the chosen course in the Hons. / Minor Degree Course.

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Max. Marks
1	COM-504	Professional Industrial Training / Internship	Summer Internship	3	0	0	0	100

Note: Students are required to complete the professional industrial training / Internship during summer semester breaks.

Semester 5

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
1	COM-501	Python Programming-I	PCC	4	3	1	0	50	100	150
2	COM-502	Computer Networks	PCC	4	3	1	0	50	100	150
3	COM-503	Microprocessors	PCC	4	3	1	0	50	100	150
4	HSMC-501	Open Elective1	OEC	3	2	1	0	50	100	150
5	COM-511	Python Programming Lab	PCC	2	0	0	4	50	0	50
6	COM-512	Computer Networks Lab	PCC	1	0	0	2	50	0	50
7	COM-513	Microprocessor Lab	PCC	1	0	0	2	50	0	50
8	HSMC-511	Placement Preparation-I	HSMC	2	0	0	4	50	0	50
Total				24*	27					900*

Note: Summer Internship Credits/Marks are also added in this semester's marksheet.

Hons. / Minor Degree Course

Course Code	Specialisation Track	Course Type	Cd	L	T	P	Marks		
							Mid-Semester	End-Semester	Total
MDC-AI/CS/CC/IoT/DS-501	Artificial Intelligence and Machine Learning Cyber Security IoT Data Science Cloud Computing	MOOC	5	5	0	0	25	125	150

*Note: Student can enroll in only one course from chosen Hons. /Minor Degree Course. Total Marks is 150 for any one of the chosen course in the Hons. / Minor Degree Course.

Semester 6

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
1	COM-601	Artificial Intelligence with Computer Vision	PCC	4	3	1	0	50	100	150
2	COM-602	Fundamentals of Blockchain Technology	PCC	4	3	1	0	50	100	150
3	COM-603	Software Engineering	PCC	4	3	1	0	50	100	150
4	COM-604	Theory of Computation	PCC	4	3	1	0	50	100	150
5	COM-611	Artificial Intelligence with Computer Vision Lab	PCC	2	0	0	4	50	0	50
6	COM-612	Mini Project Lab	PCC	1	0	0	2	50	0	50
7	HSMC-611	Placement Preparation-II	HSMC	2	0	0	4	50	0	50
8	NCC-601	MOOC	NCC	0	2	0	0	-	-	S/NS*
Total				21	29					750

Hons. / Minor Degree Course

Course Code	Specialisation Track	Course Type	Cd	L	T	P	Marks		
							Mid-Semester	End-Semester	Total
MDC-AI/CS/CC/IoT/DS-601	Artificial Intelligence and Machine Learning Cyber Security IoT Data Science Cloud Computing	MOOC	5	5	0	0	25	125	150

*Note: Student can enroll in only one course from chosen Hons. /Minor Degree Course. Total Marks is 150 for any one of the chosen course in the Hons. / Minor Degree Course.

Semester 7

Scheme-1

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Total Marks
1	COM-704	Professional Industrial Training/ Internship	Summer Internship	4	0	0	0	150

Note: Students are required to complete the professional industrial training / Internship during summer semester breaks

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
1	COM-701	Professional Elective 1	PEC	3	2	1	0	50	100	150
2	COM-702	Professional Elective 2	PEC	3	2	1	0	50	100	150
3	COM-703	Professional Elective 3* (SWAYAM/NPTEL MOOC/any other Platform)	MCC*	3	0	0	0	-	100	100
4	OEC-701	Open Elective	OEC	3	3	0	0	50	100	150
5	COM-705	Seminar	PEC	2	0	0	4	-	100	100
6	COM-711	Minor Project	PROJ	3	0	0	6	100	150	250
7	NCC-701	Essence of Indian Knowledge and Tradition	NCC	0	0	0	2	Satisfactory/ Not Satisfactory		S/NS*
			Total	21*	21					1050*

***Note:** Summer Internship Credits/Marks are also added in this semester's marks sheet.

Scheme-2

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Mid Semester	Final	Total
1	PII-701	Industry Internship-I	Professional Industry Internship	15	0	0	30	350	500	850
2	COM-703	Professional Elective 3* (SWAYAM/NPTEL MOOC/any other Platform)	MCC*	3	0	0	0	-	100	100
3	COM-706	Professional Elective 4* (SWAYAM/NPTEL MOOC/ any other platform)	MCC*	3	0	0	0	-	100	100
			Total	21*	30					1050*

Under Scheme 1: The students must opt for two Professional Electives & one open elective course out of the given Professional/Open elective courses. One of the Professional Elective courses must be a MOOC selected from the list of MOOCs mentioned in the MOOCs
list published by the department.

Under Scheme 2: Professional Elective 3*/Professional Elective 4* courses must be a MOOC selected from the list of MOOCs mentioned in the MOOCs list.

Note: The Department will publish a list of MOOCs available on SWAYAM/NPTEL/ any other platform before the commencement of the Next semester

Semester 8

Scheme-1

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
1	COM-801	Professional Elective 1	PEC	3	2	1	0	50	100	150
2	COM-802	Professional Elective 2	PEC	3	2	1	0	50	100	150
3	COM-803	Professional Elective 3* (SWAYAM/NPTEL MOOC/Any Other Platform)	MCC*	3	0	0	0	-	100	100
4	COM-811	Major Project	PROJ	6	0	0	12	200	300	500
			Total	15	18					900

*Note: One of the Professional Elective courses must be a MOOC selected from the list of MOOCs mentioned in the MOOCs list.

Department to publish list of available MOOCs on SWAYAM/NPTEL/Any Other platform before the Commencement of the Next semester.

Scheme-2

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Internal	External	Total
1	PII-801	Industry Internship-II	Professional Industry Internship	15	0	0	30	600	300	900
			Total	15	30					900

Annexure 1: Open Electives Course List

S. No.	Semester	Course Code	Course Name
1	7	OEC-701	(A) Strategic Management
2			(B) Marketing Management
3			(C) Managerial Economics

Annexure 2: Professional Elective Courses List

S. No.	Semester	Course Code	Course Name
1	7	COM-701	(A) Cloud Computing and Services
2			(B) High Performance Computing
3			(C) Soft Computing
4		COM-702	(A) Object-Oriented Analysis and Design
5			(B) Software Testing and Automation
6			(C) Applied Data Science
7	8	COM-801	(A) Advanced Machine Learning
8			(B) Generative AI
9			(C) Knowledge Representation and Reasoning
10		COM-802	(A) UAVs and Drone Technology
11			(B) Smart City Technologies
12			(C) Foundations of Internet of things and Industry 4.0

Overall Credits and Marks Distribution

Semester	Major Degree		Minor Degree / Specialization course.	
	Credits	Total Marks	Credits	Total Marks
1	22	750	0	0
2	22	800	0	0
3	26	1000	5	150
4	21	750	5	150
5	24	900	5	150
6	21	750	5	150
7	21	1050	0	0
8	15	900	0	0
Total	172	6900	20	600

Semester 1

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
BSC-101	Engineering Mathematics-I	BSC	5	4	1	0	50	100	150

Course Outcomes

At the end of the course the student will be able to	
CO1	Apply the knowledge of calculus to plot graphs of functions and solve the problem of maxima and minima.
CO2	Determine the convergence/divergence of infinite series, approximation of functions using power and Taylor's series expansion and error estimation.
CO3	Apply the concept of definite integrals to calculate area under the curves.
CO4	Understand and apply the concepts of matrices
CO5	Demonstrate knowledge of vector space by solving associated problems

Detailed Syllabus

Section-A

Unit 1: Differential Calculus I: Leibnitz theorem (without proof), Partial differentiation, Euler's theorem on homogeneous functions, Asymptotes, Double points, curvature, Curve tracing in Cartesian, polar and parametric forms. **(11 Hrs)**

Unit 2: Differential Calculus II: Rolle's Theorem, Mean value Theorem, Taylor's and Maclaurin's series with remainder, indeterminate forms, Taylor series in two variables, Maxima and Minima of functions of two variables, method of Lagrange's multipliers.

(11 Hrs)

Unit 3: Integral Calculus: Definite integrals with important properties, differentiation under the integral sign, Gamma, Beta and error functions with simple problems, applications of definite integrals to find length, area, volume and surface area of revolutions, transformation of coordinates, double and triple integrals with simple problems.

(10 Hrs)

Section-B

Unit 4: Matrices- Matrices, vectors: addition and scalar multiplication, matrix multiplication, Linear systems of equations, linear Independence, rank of a matrix, determinants, Cramer's Rule, inverse of a matrix, Gauss elimination and Gauss-Jordan elimination.

(15 Hrs)

Unit 5: Vector Space, linear dependence of vectors, basis, dimension, Linear transformations, range and kernel of a linear map, rank and nullity, Inverse of a linear transformation, rank nullity theorem, composition of linear maps, Eigenvalues, eigenvectors, symmetric, skew-symmetric, and orthogonal matrices, eigenbases. **(10 Hrs)**

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Engineering Mathematics	B.S. Grewal	Khanna Publications, New Delhi	44th (2018)
2	Calculus and Analytic Geometry	Thomas and Finney	Addison Wesley, Narosa.	1st (2010)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Advanced Engineering Mathematics	V. P. Mishra, Jyoti Sharma, Pratibha Mishra	Bhavya Books	2nd (2019)
2	Integral Calculus	S. Narayan	S. Chand, New Delhi	35th (2005)
3	Engineering Mathematics	N.P Bali	Laxmi Publications	13th (2016)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
BSC-102	Engineering Physics	BSC	5	4	1	0	50	100	150

Course Outcomes

At the end of the course the student will be able to	
CO1	Apply the concept of gradient, divergence, and curl to understand advance physics.
CO2	Use Maxwell's equations to describe propagation of electromagnetic waves in a medium.
CO3	Apply the concept of wave function to solve problems related to particle confined in a box.
CO4	Understand the concept of interference, diffraction, and polarization of light.
CO5	Understand and articulate the working principle of lasers and optical fibres

Detailed Syllabus

Section A

Unit 1: Mathematical Physics: Concepts of Del Operator; Gradient of scalar, divergence and Curl of vector, Gauss divergence theorem and Stokes theorem **(10 Hrs)**

Unit 2: Electromagnetic Theory: Displacement Current, Maxwell's equations in vacuum and non-conducting medium, Electromagnetic wave propagation in free space (EM wave equations for electric and magnetic fields for free space) and their solutions (plane wave solution), Velocity of electromagnetic waves. **(10 Hrs)**

Unit 3: Quantum Mechanics: Inadequacies of Classical Mechanics, de-Broglie's concept of matter waves, Wave-packet (Wave-group), Phase and Group velocity, Heisenberg's uncertainty Principle, Experimental illustration of Uncertainty principle using single slit.Wave function: Definition, Interpretation and its significance, Schrodinger's Wave equation (Steady state and time dependent) for one dimension case, Concept of Operators and expectation Values, Applications of Schrodinger's equation (Time Independent) to: a) Particle in a One-Dimensional Box of infinite height, b) Single Step Potential Barrier. **(16 Hrs)**

Section B

Unit 4: Applied Optics: Interference in thin films (by reflection and transmission of light), Theory of Newton's rings by reflected light, Determination of wavelength and refractive index of monochromatic light by Newton's rings theory, Fraunhofer and Fresnel's diffractions, Fraunhofer diffraction due to a single slit, Plane diffraction grating and its theory for secondary maxima and minima, Unpolarised and polarized light, Double refraction phenomenon, Nicol Prism, Mathematical representation of elliptically and circularly polarized light, Quarter and Half wave plates, Numerical problems. **(15 Hrs)**

Unit 5: Principal of Laser action, Einstein's coefficients, Ruby Lasers, Propagation of Light in Optical fibres, Acceptance angle and acceptance cone, Numerical Aperture, Single mode and Multimode fibres, Characteristics and General applications of Lasers and Optical fibres, Numerical problems. **(5 Hrs)**

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Vector Analysis	Murray R. Spiegel	McGraw Hill Education	2nd (2017)
2	Fundamentals of Physics	Robert Resnick Jearl Walker, David Halliday	Wiley	10th (2015)
3	Concepts of Modern Physics	Arthur Beiser, Shobhit Mahajan, S. Rai Choudhury	McGraw Hill Education	7th (2017)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Engineering Physics	H. K. Malik and A. K. Singh	McGraw Hill Education	2nd (2017)
2	Engineering Physics	S. Sharma and J. Sharma	Pearson India	1st (2018)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-101	Introduction to C Programming	ESC	5	4	1	0	50	100	150

Course Outcomes

At the end of the course the student will be able to	
CO1	Design flowcharts, algorithms and pseudocode for solving problems.
CO2	Understand the syntax and semantics of C programs and use them to translate the algorithms into programs.
CO3	Decompose a problem into functions and synthesize a complete program using divide and conquer approach.
CO4	Debug and test programs to evaluate program correctness.
CO5	Implement derived and user-defined data types and files in C programming for a given application.

Detailed Syllabus

Section-A

Unit 1: Introduction to Programming: Evolution of programming languages, Structured Programming, Compilation process, object code, source code, executable code, Operating systems, Fundamentals of algorithms, flow charts and pseudocodes.

(6 Hrs)

Unit 2: Introduction to C: Introduction, Importance of C, Sample C programs, Basic structure of C programs, Executing a C program, Character set, Keywords, Identifiers, Constant and Variables, Data types, Operators, Precedence of operators, Statements, Expressions, Input-output functions.

(8 Hrs)

Unit 3: Control Statements, Storage Classes, Library Functions: Control Structures: Decision making and Branching, Control Structures: Decision making and looping, Storage Classes: Types of storage class, Scoping rules, Standard Library Functions, advantages and their use (I/O functions, String, Character, Mathematics, Time and Date functions).

(12 Hrs)

Section-B

Unit 4: User-Defined Functions, Arrays, Recursion, Handling of character strings, Structures, Unions, User defined and standard functions, Formal and Actual arguments, Functions category, function prototypes, parameter passing, Call-by-value, Call-by-reference, Nested Functions, Recursion, One dimensional Array, Multidimensional Array declaration and their applications, Passing array to a function, String Manipulation, Declaration of structures, declaration of unions, pointer to structure and unions.

(12 Hrs)

Unit 5: Pointers, Dynamic memory allocation, File management in C, Pointer variable and its importance, Pointer Arithmetic, passing parameters by reference, pointer to pointer, pointer to functions, Dangling pointer, Dynamic memory allocation, Console input output functions, Disk input output functions, opening closing and creating Data files.

(10 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Programming with C	E. Balaguruswamy	McGraw Hill Education	7th (2017)
2	Programming with C	Byron Gottfried	McGraw Hill Education	4th (2018)
3	C Programming Language	Brian W. Kernighan, Dennis M. Ritchie	Pearson	2nd (2015)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	C The Complete Reference	Herbert Schildt	McGraw Hill Education	4th (2017)
2	C How to Program	Paul J. Deitel	Pearson	8th (2015)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
HSMC-101	Design Thinking	HSMC	3	2	0	0	50	100	150

Course Outcomes

At the end of the course the student will be able to	
CO1	Articulate the fundamentals of the Design Thinking Framework for problem solving.
CO2	Understand the needs and challenges of different stakeholders in the given problem domain.
CO3	Apply design thinking framework and strategies to develop innovative solutions for complex business problems.
CO4	Evaluate the financial viability of the proposed solution (product/service).
CO5	Work in a team to develop, validate and test the solution.

Detailed Syllabus

Section-A

Unit 1: Design Thinking: Introduction of Design Thinking; Framework: 5 Phase Design i.e. Empathize, Define, Ideate, Prototype, and Test; Case Study, Design Thinking in the organization.

(6 Hrs)

Unit 2: Empathize Phase: Discover design problems; Overview of the empathize phase; Define project stakeholders; Methods within empathize phase. Define Phase: Frame the design problems, Overview of the define phase, Methods within Define phase.

(8 Hrs)

Unit 3: Ideate Phase: Come up with multiple solutions, Overview of the Ideate phase, Methods within the Ideate phase.

(5 Hrs)

Section-B

Unit 4: Project Proposal: Project Identification - Developing Business Idea, Preparing Feasibility Report; Project Formulation - Feasibility Analysis Techno - Economic Analysis, Financial Analysis, Profitability Analysis; Significance of a business plan, components of a business plan.

(8 Hrs)

Unit 5: Prototype phase: Design the solutions, Overview of the Prototype phase, Methods within Prototype phase. Test phase: Validate the solutions, overview of the Test phase, methods within test phase.

(8 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Designing for growth: A design thinking tool kit for managers	Jeanne Liedtka and Tim Ogilvie	Columbia University Press	1st (2011)
2	The design thinking playbook: Mindful digital transformation of teams, products, services, businesses and ecosystems	Michael Lewrick, Patrick Link, Larry Leifer	Wiley	1st (2018)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Presumptive design: Design provocations for innovation	Leo Frishberg and Charles Lambdin	Morgan Kaufmann	1st (2016)
2	Systems thinking: Managing chaos and complexity: A platform for designing business architecture., Chapter Seven: Design Thinking	Jamshid Gharajedaghi	Morgan Kaufmann	3rd (2011)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
BSC-112	Engineering Physics Lab	BSC	1	0	0	2	50	0	50

Course Outcomes

At the end of the course the student will be able to	
CO1	Apply the concept of magnetic field to understand the working of electric vibrators and Faraday's laws.
CO2	Relate the phase variation between current and voltage through inductor, capacitor and resistor in LCR series and parallel circuits.
CO3	Measure and analyze the working of diodes and transistors in different configurations.
CO4	Measure and analyze the intensity variation of light due to interference, diffraction and polarization.
CO5	Examine the basic operation of laser, solar cell and its application i.e Planck's constant.

List of Activities for Engineering Physics Lab

S. No.	Activities
1	To measure the frequency of A.C. mains using an electrical vibrator.
2	To analyze the variation of EMF with respect to velocity of magnet to verify Faraday's laws.
3	To measure the impedance of LCR circuit
4	To verify and plot the V-I characteristics of a PN junction diode.
5	To observe the common base/ common emitter characteristics of PNP/NPN transistors.
6	To verify Zener diode characteristics
7	To determine the dispersive power of a given prism using a spectrometer.
8	To compute the wavelength of monochromatic light using Newton's rings apparatus.
9	To determine the wavelength of sodium light using a plane transmission grating
10	To determine the specific rotation of sugar/glucose using Laurent's Half shade polarimeter.
11	To evaluate the value of Planck's constant using a photo-cell.
12	To examine the characteristics of a Solar cell.
13	To determine the wavelength of a He-Ne laser.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-111	C Programming Lab	ESC	2	0	0	4	50	0	50

Course Outcomes

At the end of the course the student will be able to	
CO1	Demonstrate an understanding of the overall syntax and semantics of C programs.
CO2	Break problems into smaller solvable problem statements and integrate to create a complete solution.
CO3	Develop readable, modular and reusable C programs using built-in and user-defined functions.
CO4	Debug and test programs to determine that the program performs as expected.
CO5	Develop programs that perform operations using derived data types and files.

List of Activities for C Programming Lab

S. No.	Activities
1	Familiarization with the Lab Environment. Program to print "Hello World".
2	Program for arithmetic operations like addition, subtraction, multiplication and remainder. Program to calculate area and perimeter of circle, square and rectangle. Program to find the Euclidean distance between two points in a plane.
3	Program to check if a number is even or odd. Program to check whether the alphabet is a vowel or consonant. Program to calculate area and perimeter of circle, square and rectangle based on user's choice.
4	Program to calculate factorial of a number using a for loop. Program to print the Fibonacci sequence using for loop. Program to find maximum of n numbers using for loop.
5	Program to check if a number is prime or not using a while loop. Program to calculate sum of digits of a number using a while loop. Program to check if a number is Armstrong or not using a while loop.
6	Program to find the maximum element and index in the array. Program to sort an array using bubble sort. Program to search an element in an array using linear search.
7	Program to read and print elements in two dimensional arrays. Program to perform matrix multiplication.
8	Program to demonstrate the use of various string operations. Program to check whether a given string is a palindrome.
9	Program to create Simple Calculator using switch case and function for every operation. Program to print the Fibonacci sequence using recursion. Program to calculate factorial of a number using recursion.
11	Program to differentiate pass by value and pass by reference by swapping two numbers using function.
12	Program to demonstrate the use of malloc(), calloc(), realloc() and free() functions.
13	Program to store the information of student marks using structures and find total marks of individual student. Program to store information of a student using union. Program to implement an array of structures for student marks for each student and pass it to function for various operations.
14	Program to read and write in a file. Program for reading and writing the student marks data to files.
15	ini Project: Student Information System.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
NCC-101	Induction Training	NCC	0	0	0	4	-	-	S/NS*

Course Outcomes

At the end of the course the student will be able to	
CO1	Understand the key initiatives undertaken by the Institute and perceive the benefits of enrolling into the program.
CO2	Develop ICT skills through modern tool usage.
CO3	Develop understanding regarding different career paths and options.
CO4	Explore self-learning opportunities through MOOCs, Digital Library and online digital resources.
CO5	Leverage self-development opportunities available on campus.

List of Activities for Induction Training

S. No.	Activities
1	Orientation Session on Institution.
2	Address by the Director of the Institution.
3	Mantra to Success using ICT Tools at MIET: Deep Dive into Official Email Id, Google Drive, Google Classroom and Google Meeting.
4	Introduction to Google CS First.
5	Personality Development workshop.
6	Hands-On session on ERP system.
7	Project Showcasing Session: AI, Computer Vision and RPA.
8	E-Learning Resources at MIET (MOOC Platforms).
9	PI 360 Training session.
10	Student Clubs at MIET (Ureka, Literary).
11	Orientation session on the Library.
12	Extra-Curricular Activities in College.
13	Orientation session on Other Committees (NSS and Cultural).
14	Familiarization to Dept./Branch.
15	Orientation on Mentor Mentee Program.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
MCC-102	Indian Constitution	MCC	1	1	0	1	50	-	50

Course Outcomes

At the end of the course the student will be able to	
CO1	Understand the emergence, evolution, structure and composition of Indian Constitution.
CO2	Understand and analyze federalism in the Indian context.
CO3	Understand and analyze Panchayati Raj institutions as a medium of decentralization and the three organs of the state in the contemporary scenario.
CO4	Understand and analyze the Indian Political scenario amidst the emerging challenges.
CO5	Evaluate Indian foreign relations under the cold war and post cold war era.

Detailed Syllabus

Section-A

Unit 1: Introduction to Indian Constitution: Historical Background of the Constituent Assembly, Government of India Act of 1935 and Indian Independence Act of 1947, Composition of the Constituent Assembly, Functions of the Constituent Assembly, Various Committees of the Constituent Assembly, Enforcement of the Constitution, Indian Constitution and its Salient Features.

(5 Hrs)

Unit 2: The role of B R Ambedkar in the making of the Indian Constitution, The Preamble of the Constitution, Fundamental Rights, Fundamental Duties, Directive Principles of State Policy, Gandhian Principles, Liberal Principles, Socialistic Principles.

(5 Hrs)

Unit 3: Indian Federalism: Meaning and Definition of Federalism, Structure and Features of Indian Federalism, Difference between Indian and Federation of other states, Difference between federal and unitary features, Critical Evaluation of the Indian Federal System, Decentralisation of Powers, Centre-State Relations, 73rd Amendment, Panchayath Raj Institutions.

(5 Hrs)

Unit 4: Union Government: Powers of Indian Parliament, Functions of Rajya Sabha, Functions of Lok Sabha, Powers and Functions of the President, Comparison of powers of Indian President with the United States, Powers and Functions of the Prime Minister.

(5 Hrs)

Unit 5: India's External Relations - Cold War and Post-Cold War era: What is Foreign Policy? Basic Determinants of Foreign Policy, Indian and its Neighbours, India's Extended Neighbourhood in West Asia and South-East Asia, India's relations with the United States and Russia, India and the World Organisations, India in the 21st century.

(5 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Working of a Democratic Constitution of India	G. Austin	New Delhi: Oxford University Press	1st (2003)
2	Introduction to the Constitution of India	D. D. Basu	Lexis Nexis	24th (2019)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Contemporary India: Economy, Society, Politics	N. Chandhoke and Priyadarshini	Pearson Education India	1st (2009)
2	Understanding Contemporary India: Critical Perspectives	A. Vanaik and R. Bharghava	Orient Blackswan	1st (2010)

SEMESTER 2

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
BSC-201	Engineering Mathematics-II	BSC	5	3	2	0	50	100	150

Course Outcomes

At the end of the course the student will be able to	
CO1	To understand probability and random variables and various discrete and continuous probability distributions and their properties
CO2	Calculate probabilities, and derive the marginal and conditional distributions of bivariate random variables
CO3	Analyse statistical data using measures of central tendency, dispersion and location
CO4	Understand and discuss the issues surrounding sampling and significance
CO5	Develop analytical skills in structuring and interpreting the business problems statistically

Detailed Syllabus

Section-A

Unit 1: Probability spaces, conditional probability, independence; Discrete random variables, Independent random variables, the multinomial distribution, Poisson approximation to the binomial distribution, infinite sequences of Bernoulli trials, sums of independent random variables; Expectation of Discrete Random Variables, Moments, Variance of a sum, Correlation coefficient, Chebyshev's Inequality. (**9 Hrs)**

Unit 2: Continuous random variables and their properties, distribution functions and densities, normal, exponential and gamma densities. Bivariate distributions and their properties, distribution of sums and quotients, conditional densities, Bayes' rule.

(12 Hrs)

Unit 3: Measures of Central tendency: Moments, skewness and Kurtosis - Probability distributions. Binomial, Poisson and Normal - evaluation of statistical parameters for these three distributions, Correlation and regression – Rank correlation **(10 Hrs)**

Section-B

Unit 4: Hypothesis – Introduction, Format and Types; Procedure of Hypothesis Testing; Errors in Hypothesis; Two-tail and One-tail Test of Hypothesis; Tests of Significance for Attributes; Tests of Significance for Variables; Tests of Significance for Small Samples; t-distribution and its application

(11 Hrs)

Unit 5: Difference of means and correlation coefficients, test for ratio of variances - Chi-square test for goodness of fit and independence of attributes **(10 Hrs)**

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Advanced Engineering Mathematics	Erwin Kreyszig	Wiley	10th (2015)
2	A First Course in Probability	S. Ross	Pearson Education India	6th (2002)
3	"Introduction to Probability and Statistics for Engineers and Scientists	Sheldon M. Ross	Academic Press	5th (2009)

Reference Books

S.No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Advanced Engineering Mathematics	R.K. Jain, S. R. K. Iyenger	Narosa Publishing House Pvt. Ltd.	5th (2016)
2	Higher Engineering Mathematics	Dr. B. S. Grewal	Khanna Publications	43rd (2017)
3	Engineering Mathematics	N.P Bali	Laxmi publications	13th (2009)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-201	Data Structures using C	PCC	4	3	1	0	50	100	150

Course Outcomes

At the end of the course the student will be able to	
CO1	Understand the basics of data types and data structures.
CO2	Compare different data structures in context of their properties.
CO3	Identify the use of appropriate data structures to solve a given problem.
CO4	Apply different data structures to solve different sorting and searching problems.
CO5	Appreciate applicability of advanced data structures to model and solve real world problems.

Detailed Syllabus

Section-A

Unit 1: Introduction to data structures: Binary and Decimal Integers, Real Numbers, Character Strings, Abstract Data Types, Sequences as Value Definitions, Data Types in C, Pointers in C, Data Structures and C, Representation of Arrays, Structures in C.

(3 Hrs)

Unit 2: Stacks: Concept of Stacks, Operation on Stacks, Representing Stacks in C, Implementing the pop Operation, Testing for Exceptional Conditions, Implementing the Push Operation, Multiple stacks, Application of stacks in Infix, Postfix, and Prefix, Recursion.
Queues: Concept of Queues, Operation on Queues, Representing Stacks in C, Multiple Queues, Priority Queues, Circular Queues.

(10 Hrs)

Unit 3: Linked Lists: Concept of Linked Lists, Representing Linked Lists in C, Insertion, Deletion and Traversal on Linear Linked Lists, Doubly Linked List, Circular Linked List, Linked List as Data Structure, Header nodes, Implementation of Stacks and Queues using linked list, Dynamic memory management, Garbage Collection.

(10 Hrs)

Section-B

Unit 4: Trees: Binary trees and its representation using Linked list, Operations on Binary Trees, Traversal Algorithms, Applications, Threaded Binary Trees and its Traversal algorithms, Heterogeneous Binary Trees, List representation using Binary Trees, Optimum Search Trees, AVL trees.
Graphs: Representation of Graphs, Traversal methods, Applications Undirected Graphs, Directed Graph and Traversal, Depth First Search, Breadth First Search

(18 Hrs)

Unit 5: Sorting and Searching: Exchange Sort (Bubble, Quicksort) Selection and Tree Sorting Insertion sort, Shell Sort, Address Calculation Sort, Merge and Radix Sort, Sequential Searching, searching an Ordered Table, Index sequential search, Binary search, Interpolation search, Tree searching.

(5 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Data Structure using C	Langsam, Yedidyah, Moshe J. Augenstein, and Aaron M. Tenenbaum	Pearson Education	1st (2019)
2	Data Structures and Program Design in C	Robert L. Kruse and Bruce P. Leung	Pearson Education	2nd (2006)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Fundamentals of Data Structures	Horowiz E. and Sahni S.	Universities Press	2nd (2008)
2	An Introduction to Data Structures with Application	Jean-Paul Tremblay and Paul Sorenson	McGraw Hill Education	2nd (2017)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
ESC-201	Basic Electrical and Electronics Engineering	ESC	4	3	1	0	50	100	150

Course Outcomes

At the end of the course the student will be able to	
CO1	Learn about applications of network laws and theorems to design electric circuits.
CO2	Examine Sinusoidal waveforms and classify measuring instruments
CO3	Explain the construction and principle of operation of single-phase transformers and DC Machines.
CO4	Understand the characteristics of semiconductor devices and rectifier circuits
CO5	Interpret the principle of various transistor configurations and characteristics.

Detailed Syllabus

Section-A

Unit 1: Elementary Concepts of DC and AC Circuits: Recent advancements in Electrical Engineering, DC Circuit elements (R, L and C), Voltage and Current sources, Kirchhoff's Current and Voltage laws, mesh and nodal analysis. Superposition, Maximum power transfer, Thevenin and Norton theorems.

(8 Hrs)

Unit 2: Representation of Sinusoidal Waveforms: Peak and RMS values, Phasor representation, Real power, Reactive power, apparent power, power factor. Analysis of single-phase AC circuits. Classification of Instruments: Operating principle of Measuring Instruments. **(8 Hrs)**

Unit 3: Single phase Transformers and Machines: Principle of operation, ideal and practical transformer, equivalent circuit, losses in transformers, transformer testing, regulation and efficiency, Principle of operation of DC and AC machines.

(6 Hrs)

Section-B

Unit 4: Semiconductor Devices and Applications: Evolution in Electronics (vacuum tubes to nano-electronics), Types and specifications of Resistor, Inductor, Capacitor and Color coding, PN junction and Zener diode characteristics, Types of diodes (Tunnel diode, Schottky diode, LED, photodiode, Varactor diode), Voltage Regulators, Wave Shaping Circuits (rectifiers, filters, clippers and clampers). **(8 Hrs)**

Unit 5: Transistors and Biasing Circuits: BJT: Principle and operation of NPN transistor, configuration and characteristics (CB, CE, and CC), types of biasing circuit. Hybrid Parameters Introduction. Two port networks, hybrid model for CE, CC, CB configuration and their analysis using h-parameters, Miller theorem. FET: Principle of Operation and characteristics of JFET, biasing of FET, MOSFET and CMOS.

(10 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Basic Electrical Engineering	D. P. Kothari and l. J. Nagrath	Tata McGrawHill	4th (2019)
2	Integrated Electronics	Millman and Halkias	Tata McGraw Hill	4th (2015)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Basic Electrical Engineering	A. E. Fitzgerald, David E Higginbotham and Arvin Grabel	Tata McGraw Hill	5th (2009)
2	Electronic Devices and Circuit Theory	Boylstead	Pearson	11th (2015)
3	Electronic Principles	Malvino Leach	Tata McGraw Hill	7th (2017)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
HSMC-201	Technical Communication	HSMC	4	4	0	0	50	100	150

COURSE OUTCOME

At the end of the course the student will be able to	
CO1	Understand the constructs of written communication.
CO2	Develop competence in writing technical papers, research articles, proposals and reports.
CO3	Understand different aspects of verbal and non-verbal communication.
CO4	Gain effective communication skills and professional etiquettes for the workplace.
CO5	Demonstrate linguistic competence for public speaking and group discussion.

Detailed Syllabus

Section-A

UNIT 1: Basic Writing Skills: Sentence, Creating Coherence, Use of Phrases and Clauses in Sentences, Organizing Principles of Paragraphs in Documents, Techniques for Writing Precisely. Identifying Common Errors in Writing: Subject–Verb Agreement, Noun–Pronoun Agreement, Misplaced Modifiers, Redundancies, Clichés. The e-English: E-Mail Communication, Internet Abbreviations, Blogging, Challenges of English Language Online.

(7 Hrs)

UNIT 2: Technical Writing Skills: Letter Writing, Letter of Application Content, Format, Report Writing – Types, Structures, Data Collection, Content, Form, Writing a Proposal, Recommendations, Instructions, Business Communication. Creative Writing Skills: Free Writing, Biographical Writing, Autobiographical Writing, Process Description, Bar Charts and Flowcharts, Descriptive Writing, Argumentative Writing, Essay Writing, Précis Writing, Paraphrasing, Poster Making.

(7 Hrs)

UNIT 3: Reading and Listening: Improving Reading Skills: Skimming and Scanning, Reading and Note-Making, Intensive Reading and Predicting Content, Reading and Interpretation, Reading – Critical Reading, Hints Development. Listening Skills: Listening Comprehension, Difference between Listening and Hearing, Types of Listening, Types of Listening Intensity, Effective Listening, Ways to Improve Listening Skills, Listening and Note-Taking, Barriers to Effective Listening.

(7 Hrs)

Section-B

UNIT 4: Verbal Communication

Process of Communication and Effective Speaking: Communication Process, Barriers to Effective Communication, Flow of Organizational Communication, Language as a Tool of Communication, Pronunciation, Intonation, Stress and Rhythm, Introduction to Phonetics,

(5 Hrs)

UNIT 5: Non-Verbal Communication: Oral Communication, Communication at Workplace, Public Speaking, Persuasive Speaking, Impromptu Speaking - Extempore, Just a Minute, Debate, Conversations and Dialogues, Conversation over Telephone. Professional Etiquettes Meaning and Type, Seminar on a given topic.

(4 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Technical Communication	Wiley Editorial	Wiley	1st (2019)
2	Technical Communication: Principles and Practice	Meenakshi Raman, Sangeeta Sharma	Oxford	2nd (2011)
3	Technical Communication: A Reader-Centered Approach	Anderson	Cengage Learning	6th (2007)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-211	Data Structures using C Lab	PCC	2	0	0	4	50	0	50

COURSE OUTCOMES

At the end of the course the student will be able to: -	
CO1	Implement lists, stacks, queues and trees using arrays in C.
CO2	Create the different types of linked lists and perform its operations using C.
CO3	Create tree data structure and perform its operations using C.
CO4	Identify the data structure to develop programs for real world applications.
CO5	Assess the applicability of given data structure for a particular use-case scenario.

List of Activities for Data Structures Using C Lab

S. No.	Activities
1	Implement a List using Array and develop functions to perform insertion, deletion and linear search operations.
2	Implement a Stack using Array and develop functions to perform push and pop operations.
3	Write a program to check if a given expression is correctly parenthesized using Stacks.
4	Write a program to evaluate postfix, prefix and infix expressions using Stacks.
5	Write a program to convert an infix expression to its corresponding postfix and prefix expressions and vice-versa.
6	Implement a Queue using Array and develop functions to perform enqueue and dequeue operations.
7	Implement a Singly Linked List and develop functions to perform insertion, deletion and linear search operations.
8	Implement a Doubly Linked List and develop functions to perform insertion, deletion and linear search operations.
9	Implement a Circular Linked List and develop functions to perform insertion, deletion and linear search operations.
10	Implement a Stack using Linked List and develop functions to perform push and pop operations.
11	Implement a Queue using Linked List and develop functions to perform enqueue and dequeue operations.
12	Implement a Priority Queue using Linked List and develop functions to perform enqueue and dequeue operations.
13	Implement a Binary Tree using Array and develop functions to perform traversal, searching, insertion and deletion operations.
14	Implement a Binary Search Tree using Array and develop functions to perform traversal, searching, insertion and deletion operations.
15	Implement a Binary Tree using Linked List and develop functions to perform traversal, searching, insertion and deletion operations.
16	Implement a Binary Search Tree using Linked List and develop functions to perform traversal, searching, insertion and deletion operations.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
ESC-211	Basic Electrical and Electronics Engineering Lab	ESC	1	0	0	2	50	0	50

Course Outcomes

At the end of the course the student will be able to	
CO1	Apply fundamental concepts to solve simple DC and AC electric circuits.
CO2	Verify the basic characteristics of transformers and electrical machines.
CO3	Design diode and rectifier circuits and analyze their characteristics.
CO4	Design and evaluate various transistor biasing configurations and circuits.
CO5	Design different voltage regulators

List of Activities for Electrical and Electronics Lab

S. No.	Activities
1	Verify Characteristics of passive circuit elements (R, L, C).
2	Examine Time and frequency responses of RC, RL and RLC circuits.
3	Verify and analyze of network theorems.
4	Analyze single-phase transformers.
5	Perform the polarity test of the single phase transformer.
6	To perform open and short circuit tests on single phase transformers.
7	Measure three phase power using two Wattmeter methods.
8	Verify and Plot V-I characteristics of p-n junction and Zener diodes.
9	Verify and Plot Input and Output characteristics of BJT (CE).
10	Implement half wave and full wave rectifiers.
11	Design voltage regulator using series pass transistor.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-212	Business Process Automation with RPA Lab	ESC	2	0	0	4	50	0	50

Course Outcomes

At the end of the course the student will be able to	
CO1	Develop RPA bots for performing simple business processes using UiPath Studio.
CO2	Design RPA bots to automate data collection from multiple web sources.
CO3	Develop competencies in designing RPA bots for manipulating data across documents.
CO4	Manage the developed UiPath RPA robots using UiPath Orchestrator.
CO5	Troubleshoot problems in existing RPA bots.

List of Activities for Business Process Automation Lab

S. No.	Activity Name
1	Familiarity with UiPath Studio Academic Alliance Edition: Configure UiPath Academic Alliance Edition. Design a bot that prints “Hello World” on the screen. Design a bot to generate mathematical tables.
2	Designing basic RPA bots: Design a bot to display messages using Sequence and Flowchart activities. Design a bot to display the sum of two variables by taking two variables as input and produce the output.
3	Working with Microsoft Excel: Design a bot to open a Microsoft Excel sheet and read data into a datatable. Design a bot to compare two columns in an excel sheet. The output should display 'Match/Not Match' against the corresponding cell in the sheet.
4	Working with Desktop Automation: Design a bot to create a text file, write “Hello World” into it and save it at the desired location using the 'Recorder' feature. Design a bot for automating disk clean-up processes.
5	Working with Websites: Design a bot to scrape data from a website and store it in a Microsoft Excel sheet. Design a bot to fill a webform from the data stored in a Microsoft Excel sheet.
6	Working with Documents: Design a bot to read a true PDF file and fill the webform. Design a bot to read a word file and then create a list of unique words in an excel sheet.
7	Working with Images: Design a bot to read a scanned image of an invoice and store the extracted data in a .CSV file. Design a bot to perform Optical Character Recognition (OCR) on a saved image using OCR activities.
8	Working with Emails: Design a bot to read unread emails from inbox. Design a bot to send email with attachments to multiple receivers.
9	Handling Exceptions: Apply exception handling to previously developed automations.
10	Working with Orchestrator: Configure UiPath Orchestrator and setup tenant, machine, environment and robots. Design a bot to create a Queue in Orchestrator and store the subject of the email in .CSV. Configure Orchestrator to run a process as per a given schedule.
11	Mini Project (choose one): RPA bot to send notifications to students when their attendance falls below 75%. RPA bot to extract and collate job openings from multiple job portals. RPA bot to extract and display news from Google News.
12	Deploy Mini Project using Orchestrator.

							Marks

Course Code	Course Name	Course Type	Cd	L	T	P	Sessional	Final Exam	Total
NCC-201	Environment and Sustainability	NCC	0	2	0	0	-	-	S/NS*

*Note: S=Satisfactory, NS=Not Satisfactory

Course Outcomes

At the end of the course the student will be able to :	
CO1	Describe the relationship between Humans, Environment and Sustainability.
CO2	Articulate different environmental risks and issues and potential interventions to tackle them.
CO3	Appraise sustainable energy systems through case-studies and real-world examples.
CO4	Articulate Sustainable Infrastructure Development plan.
CO5	Appreciate global sustainability best practices in diverse domains.

Detailed Syllabus

Section-A

Unit 1: Introduction to Sustainability: Humanity and the Environment: What is Sustainability? The IPAT Equation, Human Consumption Patterns and the "Rebound" Effect, Challenges for Sustainability.
Climate and Global Change: Climate Processes; External and Internal Controls, Milankovitch Cycles and the Climate of the Quaternary, Modern Climate Change, Climate Projections.

(3 Hrs)

Unit 2: Biosphere: Introduction, Biogeochemical Cycles and the Flow of Energy in the Earth System. Biodiversity, Species Loss, and Ecosystem Function. Soil and Sustainability.
Physical Resources: Water, Pollution, and Minerals. Water Cycle and Fresh Water Supply. Water Pollution. Mineral Resources: Formation, Mining, Environmental Impact.

(4 Hrs)

Unit 3: Environmental and Resource Economics: Tragedy of the Commons. Environmental Valuation. Evaluating Projects and Policies. Solutions: Property Rights, Regulations, and Incentive Policies.
Modern Environmental Management: Systems of Waste Management. Case Study: Electronic Waste and Extended Producer Responsibility. Government and Laws on the Environment. Risk Assessment Methodology for Conventional and Alternative Sustainability Options.

(4 Hrs)

Section-B

Unit 4: Sustainable Energy Systems: Environmental Challenges in Energy, Carbon Dioxide, Air, Water and Land Use. Energy Sources and Carriers. Electricity. Energy Uses. Applications of Phase Change Materials for Sustainable Energy. Problem-Solving, Metrics, and Tools for Sustainability.

(6 Hrs)

Unit 5: Sustainable Infrastructure: The Sustainable City. Sustainability and Buildings. Sustainable Energy Practices: Climate Action Planning. Sustainable Transportation: Accessibility, Mobility, and Derived Demand. Sustainable Stormwater Management.

(4 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Sustainability: A Comprehensive Foundation	Tom Theis and Jonathan Tomkin	Open Textbook Library	1st (2015)
2	Energy, Environment, and Sustainability with MindTap	Saeed Moaveni	Cengage India Private Limited	1st (2012)
3	Improving the Sustainable Development Goals: Strategies and the Governance Challenge (Routledge Focus on Environment and Sustainability)	Lars Niklasson	Routledge	1st (2019)

Reference Book

S. No.	Name of the Book	Author	Publisher	Edition (Pub. Yr.)
1	Global Challenges to CSR and Sustainable Development: Root Causes and Evidence from Case Studies (CSR, Sustainability, Ethics and Governance)	Stephen Vertigans, Samuel O. Idowu	Springer	1st (2021)

Semester 3

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-301	Design and Analysis of Algorithms	PCC	5	4	1	0	50	100	150

Course Outcomes

At the end of the course the student will be able to	
CO1	Understand written algorithms in terms of their composite steps and transformations.
CO2	Analyze time and space complexity of well known algorithms.
CO3	Understand various algorithm design techniques.
CO4	Understand P, NP and NP-Complete algorithms and their characteristics.
CO5	Gain understanding of applicability of algorithms in devising optimal solutions to given problems in diverse domains.

Detailed Syllabus

Section-A

Unit 1: Mathematical preliminaries, time complexity and space complexity, worst-case and average-case analyses, use of order notations and related results, divide and conquer recurrences, recurrence relations: substitution method, recurrence trees, Master's theorem and its applications.

(12 Hrs)

Unit 2: Quick Sort and its analyses, Merge Sort recurrence, Strassen's matrix multiplication, fast multiplication of large integers, binary search trees, priority queues, Heaps and Heap Sort.

(6 Hrs)

Unit 3: Data structures for disjoint sets, Path compression, union by rank, Prim's and Kruskal's algorithms, Huffman coding, LZW coding, shortest paths, greedy activity selection, set cover and greedy heuristics.

(12 Hrs)

Section-B

Unit 4: Dynamic Programming basics, matrix-chain multiplication, DP solution for traveling salesman and 0/1 Knapsack problems, least common subsequences, independent sets and backtracking algorithm, Breadth-First and Depth-First search algorithms.

(12 Hrs)

Unit 5: Topological sort, recursive graph algorithms, string matching: KMP algorithm, Rabin-Karp algorithm, number theory algorithms: basics, GCD and extended Euclidean algorithm, primality testing. Non-Deterministic Algorithms, Polynomial Time Algorithms, NP-hard and NP-complete classes, Cook's Theorem, Introduction to Approximation Algorithms.

(12 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Introduction to Algorithms	T.Cormen, C. Lieserson, R.Rivest, C.Stein	Prentice-Hall/India	3rd (2009)
2	Algorithms	S. Dasgupta, C. Papadimitriou, Umesh Vazirani	McGraw Hill Education	1st (2017)

Reference Book

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Fundamentals of Computer Algorithms	Ellis Horowitz, Sartaj Sahni	Universities Press	2nd (2008)

2	Algorithms Design: Foundations, Analysis and Internet Examples	Michael T. Goodrich, Roberto Tamassia	Wiley	1st (2006)
3	The Algorithm Design Manual	Steven S Skiena	Springer	2nd (2008)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
ESC-301	Digital Electronics	ESC	4	3	1	0	50	100	150

Course Outcomes

At the end of the course the student will be able to	
CO1	Understand and examine various number systems to be used in digital design.
CO2	Minimize the expressions using Karnaugh map method and implement them using Logic Gates.
CO3	Design and analyze various combinational and sequential circuits.
CO4	Understand and analyze various analog and digital converters.
CO5	Formulate problems and simplify with state minimizing techniques.

Detailed Syllabus

Section-A

Unit 1: Data and number systems: Binary, Octal and Hexadecimal representation and their conversions; BCD, ASCII, EBDIC, Gray codes, Excess-3 and their conversions; Signed and unsigned binary number representation with 1's and 2's complement methods, Binary arithmetic, overview of boolean algebra, simplification of logic variable using K-map method, Quine McCluskey method.

(12 Hrs)

Unit 2: Combinational circuits: Encoder, Decoder, Comparator, Binary Adder-Subtractor, Parallel Adder, Multiplexer, De-Multiplexer, Code converters and Parity Generator and checker, Memory Systems: RAM, ROM, EPROM, EEROM, Design of combinational circuits-using ROM, Programming logic devices and gate arrays (PLAs and PLDs).

(12 Hrs)

Unit 3: Sequential Circuits: Basic memory elements, S-R, J-K (master-slave), D and T Flip-Flops, their conversions with excitation table. Various types of Registers, Synchronous and Asynchronous Counters and their design. State Machine Design, State machine as a sequential controller, Moore and Mealy state machines.

(12 Hrs)

Section-B

Unit 4: Analog and Digital Converters: Different types of A/D (flash, successive approximation, delta encoded) and D/A (R-2R Ladder and binary weighted) conversion techniques.

(6 Hrs)

Unit 5: Logic families: TTL, ECL, MOS and CMOS, their operation and specifications.

(4 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Modern Digital Electronics	R. P. Jain	McGraw-Hill	4th (2010)
2	Fundamentals of Digital Circuits	Anand Kumar	PHI	4th (2016)
3	Digital Electronics	A. K. Maini	Wiley India	1st (2007)
4	Digital Electronics	Kharate	Oxford	1st (2012)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Digital Design	M. Morris Mano and Michael D Ciletti	Pearson	5th (2012)
2	Computer System Architecture	M. M. Mano	Pearson	3rd (2012)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-302	Operating Systems	PCC	4	3	1	0	50	100	150

Course Outcomes

At the end of the course the student will be able to:	
CO1	Demonstrate understanding of the concepts, structure and design of operating systems.
CO2	Articulate the general architecture of modern computer operating systems including its impact on application design and performance.
CO3	Understand and analyze complex design choices and implementation details of: processes, resource control, physical and virtual memory, scheduling, I/O and files.
CO4	Develop understanding of inter process communication and synchronization mechanisms.
CO5	Analyze the interplay and conflicts in resource usage in a multi-user, multi-tasking environment with an understanding of the trade-offs involved.

Detailed Syllabus

Section-A

Unit 1: Introduction Concepts: Operating System functions and characteristics, Historical evolution of O.S., O.S. Services, User O.S. Interface, Computer System Architecture, O.S. Design, Implementation and structure, System calls, System Programs, Virtual Machines, Spooling.

(4 hours)

Unit 2: Process Management: Study of state models, process Scheduling, Job Scheduling, Scheduling Criteria, Scheduling Algorithms, Multiple Process Scheduling.

(6 hours)

Unit 3: Process Coordination: Synchronization: Race-Conditions, critical–Section problems, semaphores, Bounded-Buffer Problem, Readers-writers Problem, Dining –Philosophers Problem.
Deadlocks: Characteristics, Deadlock Prevention, Avoidance, Detection and Recovery.

(12 hours)

Section-B

Unit 4: Memory Management: Logical and Physical Address space, Contiguous and Non-Contiguous Memory Allocation, Paging, Structure of Page Table, Segmentation, Demand paged memory management, Page replacement, Allocation of Frames, Thrashing, Swapping and Overlays, Cache Memory.

(12 hours)

Unit 5: File Systems and Disk Storage: Files: file concept, file structure, types, access methods, directory structure, allocation methods (contiguous, linked, and indexed), free-space management (bit vector, linked list, grouping),
Disk Structure, Disk Scheduling, Disk Management, Disk Formatting, Swap Space Management, RAID Structure.

(6 hours)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Operating System Concepts	Abraham Silberschatz, Peter B. Galvin, Gerg Gagne	Wiley	9th (2015)

Reference Books

S.No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Operating System	H. M. Deitel	Pearson	3rd (2007)

2	Schaum's Outline Of Operating Systems	J. Archer Harris	McGraw Hill	1st (2019)
3	Operating System Design and Implementation	Andrew S. Tanenbaum	Pearson Education	3rd (2006)
4	Operating Systems: Internals and Design Principles	William Stallings	Pearson Education	9th (2018)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
BSC-301	Discrete Mathematics	BSC	3	2	1	0	50	100	150

Course Outcomes

At the end of the course students will be able to:	
CO1	Understand basic principles of sets and operations in sets.
CO2	Analyze relations and functions and be able to determine their properties.
CO3	Apply logical notation to describe an argument.
CO4	Evaluate the basics concepts of groups, its examples and related results.
CO5	Create graphs and trees using different transversal methods.

Detailed Syllabus

Section-A

Unit 1: Sets, Relation and Function: Operations and Laws of Sets, Cartesian Products, Finite and infinite sets, countable and uncountable sets, Binary Relation and its types, Functions and its types, Principles of Mathematical induction, Principle of inclusion and exclusion, pigeon-hole principle.

(10Hrs)

Unit 2: Algebraic Structure: Groups and sub groups, related theorems, Cosets, Normal subgroups and Group homomorphism. Rings, Integral domains and fields: examples and related results.

(8 Hrs)

Unit 3: Basic Logic: Propositional logic, Logical connectives, Truth tables, Normal forms (conjunctive and disjunctive), Validity of well-formed formula, Propositional inference rules (concepts of modus ponens and modus tollens), Predicate logic, Universal and existential quantification.

(8 Hrs)

Section-B

Unit 4: Graphs: Basic terminology, multi-graphs and weighted graphs, connectivity, walk and path, circuits and cycles, shortest path in weighted graphs, Algorithm of shortest path. Hamiltonian and Eulerian paths and circuits,
Eulerian graphs, Hamiltonian graphs, Konigsberg bridge problem, Chinese postman problem, Travelling salesperson problem, Planar graph and Euler's formula.

(11 Hrs)

Unit 5: Trees and cut sets: Trees, rooted trees, path lengths in rooted trees, Spanning trees and cut sets.

(3 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Discrete Mathematics	Kenneth Rosen	McGraw Hill Education	7th (2017)
2	Graph Theory with applications to Engineering and Computer Science	Narsingh Deo	Prentice Hall	1st (2016)
3	Discrete Mathematics structure with applications to Computer Science	Jean-Paul Tremblay and R Manohar	McGraw Hill Education	1st (2017)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Concrete Mathematics	Ronald Graham, Donald Knuth, and Oren Patashnik	Pearson Education Publishers	2nd (2008)

2	Elements of Discrete Mathematics	C. L Liu	McGraw-Hill Inc	2nd (1985)
3	Discrete Mathematics Schaum‘s Outlines	Lipschutz, S. and Mark Lipson	Tata McGraw Hill Pub. Co. Ltd., New Delhi	3rd (2007)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
HSMC-301	Principles of Management	HSMC	2	2	0	0	50	100	150

Course Outcomes

At the end of the course the student will be able to	
CO1	Understand the various approaches Management Functions: Planning, Organizing, Staffing, Leading and Controlling.
CO2	Discuss the various types of Organisation Structures.
CO3	Appreciate the importance of leadership and motivation in organizations
CO4	Analyze the feedback and control procedures in organizational setting
CO5	Demonstrate an understanding of current and relevant management knowledge.

Detailed Syllabus

Section-A

Unit 1: Management: Definition, nature, purpose and scope of management, Skills and roles of a Manager, functions, principles; Evolution of Management Thought, Scientific Management, Management by Objectives: Management by exception; McKinsey's 7-S Approach.

(5 Hrs)

Unit 2: Nature, purpose and framework of Planning: types of plans. The nature of objectives and MBO; Nature and purpose of organizing; Determinants of organization structure; Line and staff concept, New approaches in organization design; Downsizing; Span of management; Authority relationships; Delegation and decentralization.

(6 Hrs)

Unit 3: Staffing: Human Resource Management and Selection, Performance appraisal and Career strategy, Managing Change. Leading: Human Factors and Motivation, Leadership, Communication, Teams and TeamWork.

(5 Hrs)

Section-B

Unit 4: Direction as a management process: its significance and main elements; Coordination-its importance and techniques; Controlling: Concept, planning-control relationship, process of control, Types of Control, Control Techniques.

(6 Hrs)

Unit 5: Knowledge management; Total quality management; Business process re-engineering; New people management; management of productivity; Corporate Governance; Creativity and Innovation; Theory of constraints: Issues and concerns.

(3 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Principles of Management	L. M. Prasad	Sultan	10th (2019)
2	Organizational Behaviour	Stephen P. Robbins	Prentice-hall	18th (2018)
3	Essential of Management.	Koontz O'Donnel	Tata McGraw Hill Education	10th (2015)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Principles of Management	Tripathi, P.C.and Reddy, P.N.	Tata Mc Graw Hill	6th (2017)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
ESC-311	Digital Electronics Lab	ESC	2	0	0	4	50	0	50

Course Outcomes

At the end of the course the student will be able to:	
CO1	Implement and verify Boolean expressions using Logic Gates.
CO2	Apply Boolean laws and K-map to simplify the digital circuits.
CO3	Design and implement various combinational circuits using digital ICs.
CO4	Design seven segment decoder using Logical Gates.
CO5	Design and implement various sequential circuits using digital ICs.

List of Activities for Digital Electronics Lab

S. No.	Activity Title
1	To study and verify the truth table of logic gates using Logisim and Digital Kit.
2	To study and verify the truth table of logic gates using Breadboard.
3	Simulation of basic gates using Universal Gates.
4	Verify the equivalence of various Boolean expressions using Logisim and Digital Kit.
5	Verification of truth tables of ADDER and Subtractor.
6	To verify the truth table of MUX and DEMUX using NAND.
7	Verification of truth tables of BCD-7 Segment Display.
8	Truth table verification of Flip-Flops.
9	Design of Ring counter/ Johnson counter, Asynchronous Counter and Synchronous Counter.
10	To verify the truth table of one bit comparators using logic gates.
11	**Mini Project 1:** Digital stopwatch.
12	**Mini Project 2:** Water level controller.
13	**Mini Project 3:** 3 to 8 decoder along with seven segment LEDs.
14	**Mini Project 4:** Digital clock and timer circuit.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-312	Operating Systems Lab	PCC	2	0	0	4	50	0	50

Course Outcomes

At the end of the course the student will be able to	
CO1	Develop competence in shell scripting on Linux operating system.
CO2	Use system commands to manipulate files, determine and control resource usage and perform administrative tasks.
CO3	Understand the architecture of the Linux operating system and its different subsystems.
CO4	Develop programs making use of linux system calls to solve given problems.
CO5	Program complex inter-process interactions involving communication and synchronization.

List of Activities for Operating Systems Lab (C)

S. No.	Activity Title
1	● Practice basic bash, IO redirection, pipe, file and directory operations, searching files, process management and vi/nano editor commands.
2	● Use the date and who commands, in one line, such that the output of date is displayed on the screen and the output of who is redirected to a file. ● Write a sed command that swaps the first and second words in each line in a file.
3	● Write a shell script that takes a command line argument and reports on whether it is a directory or a file. ● Write a shell script that takes file names as arguments and convert all of them to uppercase.
4	● Write a shell script that determines the period for which a specified user is working on the system. ● Write a shell script that displays all the lines between start and end line numbers passed as argument. ● Write a shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.
5	Write a shell script and C program to perform the following string operations: ● To extract a substring from a given string. ● To find the length of a given string. Compare the running time of above shell script and C program using the time command.
6	Write a C program that takes, as a command line argument, the number of megabytes of memory it will use and during execution it should consume that much memory. Observe memory usage during program execution using free command.
7	Write a CPU bound C program and a I/O bound C program and observe the effect of their CPU share using the top command and its variants.
8	Observe and understand the process map of the previous two lab programs using pmap command.
9	Write a program in C that creates a child process, waits for the termination of the child and lists its PID.
10	Write a C program to print the address of a variable and enter a long loop (say using while(1)). Start three to four processes of the same program and observe the printed address values.
11	● Write C Program to demonstrate wait and signal operations in semaphores. ● Implement consumer-producer problem using semaphores.
12	Implement pipe processing using IPC.
13	Develop applications in IPC using shared memory concept and message queues.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
HSMC-312	Placement Overview and Career Planning	HSMC	1	0	0	2	50	0	50

Course Outcomes

After completion of this course the student will be able to	
CO1	Demonstrate proficiency in quantitative problem solving.
CO2	Reason logically and perform deductions on the given problem statement.
CO3	Comprehend and assimilate a variety of technical and non technical literature.
CO4	Communicate effectively for a wide variety of purposes and audiences.
CO5	Solve problems effectively in competitive exams.

List of Activities for Placement overview and Career Planning

S. No.	Activity Title
1	Practice Quantitative Aptitude: Vedic Mathematics concepts, Simplification, Quadratic Equations, Surds and Indices, Average and Weighted average, Mixtures and Allegations. Percentages, Profit and Loss, Simple and Compound Interest, Ratio and Proportion, Partnership, Age Problems, Word Problems. Time and Work; Time, speed and Distance; Mensuration- Cylinder, Cone, Sphere; Number Systems; Sequence and Series; Permutation and Combination; Probability; Logarithms; Functions; Data Interpretation. Data Arrangement, Clocks and Calendars, Different patterns of Puzzles – Quantitative Based, Data Sufficiency, order and Ranking, Direction sense.
2	Practice Logical Reasoning questions on following topics: Venn Diagram, Syllogisms / Deductions, Abstract Reasoning, Logical Connectives, Input Output, Attention to detail, Selection Decision table, Logical Sequence Words, Coding Decoding, Coded Inequalities, Number Series, Alpha Series, Analogy, Cryptarithmetic, Blood Relations.
3	Verbal Ability: General Grammar Rules and Grammar Essentials: Nouns, Verbs and Pronouns; Subject-Verb Agreement; Pronoun-Antecedent Agreement; Punctuations. Concept Building for: Reading Comprehension, Passage Analysis, Vocabulary building, Confusing words, Sentence Completion, Verbal Analogy, Spotting errors in sentences. Jumbled Sentences, Sentence Improvisation, Writing Skills - Essay Writing, Writing Skills - Email Writing, Critical Reasoning, Reading Comprehension Advanced + Para Jumbles, Gap - Bridging.
4	Learn various effective communication skills: Word Power enhancement; Attention to Detail: spacing, punctuation, spelling, and other finer aspects; Verbal and non-verbal communication skills; Written English with emphasis on writing grammatically correct technical / official letters, applications and reports; Getting rid of inhibitions and building confidence; Assertive and Submissive communication; Using Language for Convincing and Persuasion; Art of asking Questions.
5	To participate in a national level aptitude test to assess the learners ability on quantitative, verbal, logical reasoning, soft skills and technical skills.
6	To prepare for group discussion and personal interviews and understanding your personality type.
7	Explore job roles and responsibilities of different job postings and prepare your resume accordingly.
8	Explore higher education opportunities in India and abroad and shortlist the possible universities.

Hons./Minor Degree Course

Course Code	Specialization Track	Course Type	Cd	L	T	P	Marks		
							Mid-Semester	End-Semester	Total
MDC-AI/CS/CC/IoT /DS-301	Artificial Intelligence and Machine Learning/ Cyber Security/ IoT/ Data Science/ Cloud Computing	MOOC	5	5	0	0	25	125	150

The students shall be required to complete a MOOC for duration 8-12 weeks corresponding to the opted specialization track. Such MOOCs shall be approved by the Departmental Academic Committee and updated from time to time.

The Departmental Academic Committee shall assess the student work based on the following criteria:

1. Submission of course completion certificate- **75 Marks**
2. Mid Semester Test - **25 Marks**
3. End- Semester Test- **50 Marks**

SEMESTER 4

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-401	Object Oriented Programming using C++	PCC	4	3	1	0	50	100	150*

*Note: There will be unit wise online assessment conducted by IBM. Marks will be normalized for each assessment to calculate marks for the final exam. The sessional exam will comprise of presentation/viva.

Course Outcomes

At the end of the course the student will be able to:	
CO1	Appreciate the difference between structured programming and object-oriented programming.
CO2	Understand the basic concepts of objects, classes and their relationships.
CO3	Apply object-oriented decomposition to model real-world scenarios and applications.
CO4	Assimilate and implement object-oriented features in C++.
CO5	Articulate advanced features in C++ like exception handling and templates.

Detailed Syllabus

Section-A

Unit 1: Objects and Classes: Introduction to Procedural and Object Oriented Programming Languages, Structure in C and C++, Class specification, Objects, Data hiding, Encapsulation and abstraction, namespaces, Array of objects, Passing objects as arguments, Returning object from a function, inline functions, Static data member and member function, 'const' member function.

(5 Hrs)

Unit 2: Constructor and Destructor: Constructors, Parameterized Constructors, Constructor Overloading, Constructors in array of objects, Constructors with default arguments, Dynamic Initialization, Pointer to objects, this pointer, Dynamic memory allocation, Array of pointer to objects, Copy Constructor, Static objects, Friend function, and Friend classes.

(8 Hrs)

Unit 3: Operator Overloading and Type Conversion: Syntax of operator overloading, Overloading Unary and Binary operator, overloading arithmetic operator, relational operator, Overloading Unary and Binary operator using friend function, Data conversion and Overloading some special operators like (), [].

(8 Hrs)

Section-B

Unit 4: Inheritance and Polymorphism: Derived Class declaration, Public, Private and Protected Inheritance, friend function and Inheritance, Overriding member function, Forms of Inheritance, virtual base class, Abstract class, Constructor and Inheritance, Destructor and Inheritance, Advantage and disadvantage of Inheritance.
Classification of Polymorphism, Compile time and Run time Polymorphism, Pointers to derived class object, Virtual functions, Pure virtual functions. File handling: Formatted I/O, Hierarchy of file stream classes, Opening and closing a file, Working with multiple files, file modes, file pointers, Text vs Binary Files.

(12 Hrs)

Unit 5: Templates, Exception Handling and Standard Template Library: Need of template, Function templates, Function template with non-type parameter, Overloading function templates, Class templates, Class template with non-type parameter. Exception handling mechanism, Multiple Catch Blocks, Catch All exceptions, Throw an exception, Exception Specification. Fundamental idea about string, iterators, hashes and other types, The String and Vector classes vs C-style pointers.

(9 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	C++ The Complete Reference	Herbert Schildt	Tata Mcgraw Hill	4th (2003)
2	C++ Primer	Lippman B. S., Lajoie, Moo E. B.	Addison-Wesley Professional	5th (2013)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Object-Oriented Programming in C++	Lafore R.	Pearson Education	4th (2002)
2	The C++ programming language	Stroustrup B.	Pearson Education	4th (2013)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-402	Relational Database Management System (RDBMS)	PCC	4	3	1	0	50	100	150

Course Outcomes

At the end of the course the student will be able to:	
CO1	Identify the basic concepts, architecture and various data models used in Database Management Systems (DBMS).
CO2	Identify basic database storage structures and access techniques such as file organizations, indexing methods.
CO3	Design queries using Structured Query Language (SQL) for database definition and database manipulation.
CO4	Recognize the use of normalization and functional dependencies in DBMS.
CO5	Apply and relate the concept of transaction, concurrency control and recovery in DBMS

Detailed Syllabus

Section-A

Unit 1: Basic Concepts: Introduction, characteristics, history and applications, Components of DBMS, Advantages and Disadvantages of DBMS. Language and Architecture: Data Modeling, Records and Files, Abstraction and data Integration, Views, Data Independence, Data Associations, Data Models Classification.

(8 Hrs)

Unit 2: Entity Relationship Model: Basic concepts, constraints, design issues, Entity Relationship diagram, Weak Entity sets, Extended ER features, Design of ER database schema, Reduction of ER schema to tables. Relational Model: Attributes and domains, Tuples, Relations and Schemas, Relation representation, keys, Integrity Rules, Relational algebra, Relational Calculus, Data Manipulation using SQL.

(12 Hrs)

Unit 3: Relational Database Design: Normalization using Functional Dependency, Normalization using Join dependencies, Domain key normal form.

(8 Hrs)

Section-B

Unit 4: Transactions: Introduction to transaction and system concept, transaction state, desirable properties of transactions (ACID properties), Concurrent executions, Serializability, Recoverability, implementation of isolation, transaction definition in SQL. Concurrency Control: Lock based protocols, Timestamp-based protocols, Validation-based protocols, Multiple Granularity, Multi version Schemes, Deadlock Handling, Insert and Delete operations.

(12 Hrs)

Unit 5: Recovery Systems: Failure classification, Storage Structure, Recovery and Atomicity, Log based recovery, Shadow Paging, Recovery with Concurrent Transitions, Buffer Management. **(8 Hrs)**

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Database System Concepts	Korth, Silberchatz	Mcgraw Hill Education	6th (2013)

Reference Book

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Fundamentals of Database System	ElmasriRame, NavatheShamkant	Pearson Education	7th (2015)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-403	Computer Organization and Architecture	PCC	5	3	2	0	50	100	150

Course Outcomes

At the end of the course the student will be able to	
CO1	Understand the structure, including logical and functional components of a computer.
CO2	Illustrate various elementary concepts of computer architecture including syntax of register transfer language, micro-operations, instruction cycle, and control unit.
CO3	Develop understanding on techniques involved in the computer arithmetic operations.
CO4	Comprehend various instruction formats and addressing modes.
CO5	Articulate the concept of pipelining, multiprocessors, and input-output organization.

Detailed Syllabus

Section-A

Unit 1: Introduction: Overview of Digital Fundamentals.

(5Hrs)

Unit 2: Register Transfer and Micro operation: Register Transfer Language, Register Transfer, Bus and Memory Transfer, Arithmetic Micro operations, Logic Micro operations and Shift, Micro operations.

(10Hrs)

Unit 3: Basic Computer Organization and Design: Instruction Codes, Computer Registers, Computer Instructions, Timing and Control, Instruction Cycle, Memory Reference Instructions, Input-OutputInterrupts, Design of Basic Computer, Design of Accumulator Logic.

(10Hrs)

Section-B

Unit 4: Micro-programmed Control Unit: Control Memory, Address Sequencing. Central Processing Unit: Introduction, General Register Organization, Stack Organization, Instruction Formats, Addressing Modes.Computer Arithmetic Introduction, Addition and Subtraction, Multiplication Algorithms, Division Algorithms, Floating Point Arithmetic Operation, Decimal Arithmetic Unit, Decimal Arithmetic Operations.

(12 Hrs)

Unit 5: Peripheral devices, Input - Output interface, Asynchronous Data Transfer, Modes of Data Transfer, Priority Interrupt, Direct Memory Access, Input - Output Processor. Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Virtual Memory, Memory Management Hardware, Flynn's classification of parallel processing systems, pipelining concepts.

(12 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Computer System and Architecture	Mano,Morris	Pearson Education	3rd (2017)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
2	Computer Organization and Architecture	Stallings, William	Pearson Education	10th (2016)

3	Computer Organization	Carl Hamacher, ZvonkoVranesic, SafwatZaky	McGraw Hill Education	6th (2011)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
HSMC-401	Entrepreneurship and Business Strategies	HSMC	3	2	1	0	50	100	150

Course Outcomes

After completion of this course the student will be able to	
CO1	Understand the importance of entrepreneurship as a career option.
CO2	Analyze the types of entrepreneurship as well as the players in the ecosystem.
CO3	Understand different types of business organisations.
CO4	Understand the stages of the entrepreneurship development process and the resources needed for the successful development of entrepreneurial ventures.
CO5	Formulate a business development plan for a given idea.

Detailed Syllabus

Section-A

Unit 1: Entrepreneurship: Definition and Concept, Entrepreneurial Traits, Characteristics and Skills, Role of Innovation in Entrepreneurship,Types of Entrepreneurs, Entrepreneurship and Economic Development, Factors influencing Entrepreneurship, Myths and realities of entrepreneurship.

(6 Hrs)

Unit 2: Entrepreneurship and Intrapreneurship: Similarities and variance, India's start up revolution–Trends, Imperatives, benefits, the players involved in the ecosystem, Business Incubators, Rural entrepreneurship, Social Entrepreneurship, Women Entrepreneurs, Entrepreneurial success stories.

(5 Hrs)

Unit 3: Legal Forms of Industrial Ownership:Sole Proprietorship, Partnership, Joint Stock Company (Features, Merits and Demerits).

(5 Hrs)

Section-B

Unit 4: Project Proposal: Project Identification,Developing Business Idea, Preparing Feasibility Report, Project Formulation Feasibility Analysis Techno-Economic Analysis, Financial Analysis, Profitability Analysis, Social Cost Benefit Meaning,Significance of a business plan, components of a business plan, Business Excellence Model.

(7 Hrs)

Unit 5: Small Scale Industries and policies for entrepreneurship development**:** Definition of small-scale industries, objectives. Role of SSI in economic Development of India. SSI registration, NOC from Pollution Board, Machinery and equipment selection, Schemes and Policies for entrepreneurship development.

(7 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Fundamentals of Entrepreneurship	H. Nandan	Prentice Hall India	3rd (2013)
2	Business model generation	Alexander Osterwalder andYves Pigneur	Wiley	1st (2010)
3	Small scale industries and Entrepreneurship	Vasant Desai	Himalaya Publishing House	9th (2017)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Management of small scale Industries	Vasant Desai	Himalaya Publishing House	1st (2015)
2	Entrepreneurial Development	Khanka S.S.	S Chand and Company	1st (2006)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-411	Object Oriented Programming using C++ Lab	PCC	2	0	0	4	50	0	50

Course Outcomes

After completion of this course the student will be able to	
CO1	Decompose a given problem statement into classes.
CO2	Model class relationships and dependencies for real world problems.
CO3	Demonstrate the use of object oriented features in C++ to solve specific problems.
CO4	Work with files, I/O streams and external libraries in C++.
CO5	Apply standard coding conventions and techniques for debugging C++ code.

List of Activities for Object Oriented Programming using C++ Lab

S.No.	Activities
1	● Analyze and design classes for a student management system. ● To create a student class, its constructor, destructor and member functions. ● To create a course class, its constructor, destructor and member functions.
2	● To allocate appropriate access specifiers to data members of student and course class along with justification in comments. ● To add the details of a student using a parameterized constructor of student class.
3	● To update the details of a student using a member function of student class. ● To delete the details of a student using a member function of student class.
4	● To search student details by roll no using a member function of student class. ● To get the results of students by roll no using a member function of student class.
5	● To add the details of a course using a parameterized constructor of course class. ● To update the details of a course using a member function of course class.
6	● To delete the details of a course using a member function of course class. ● To get the details of a course by course_id using a member function of course class.
7	● To implement inheritance for a UG Course class, PG Course class and Diploma Course class. ● To implement polymorphism for member functions of UG Course class, PG Course class and Diploma Course class.
8	● To store student records in persistent storage i.e. files. ● To search student records using files.
9	● To edit student records using files. ● To delete student records using files.
10	● To create course records using files. ● To search records by course id using files.
11	● To edit course records using files. ● To delete course records using files.
12	● To make use of function templates to search a student by name or roll number. ● To update all the input methods in student and course classes with validations using try-catch. ● To run and debug the project on the online GDB compiler.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-412	Relational Database Management System Lab	PCC	2	0	0	4	50	0	50

Course Outcomes

After completion of this course the student will be able to	
CO1	Use ER Modelling to create data models for real world applications.
CO2	Convert ER Model to its corresponding database schema.
CO3	Create, retrieve, update and delete data using Structured Query Language (SQL).
CO4	Apply normalization techniques to optimize database design.
CO5	Demonstrate use of database concepts of transactions, cursors, stored procedures and triggers.

Lab Activities for RDBMS Lab

S. No.	Activities
1	To design the ER model of a university database and implement primary and foreign keys, integrity rules and relationships.
2	To create the university database from the ER model using Data Definition Language (DDL) in SQL.
3	To apply Data Manipulation Language (DML) to add and modify data in the university database.
4	To apply Data Control Language (DCL) commands to grant and revoke access on the university database.
5	To apply built-in functions of SQL to analyze the university database.
6	To perform grouping, ordering and aggregation operations.
7	To perform queries from multiple tables using joins and views.
8	To perform nested queries from multiple tables.
9	To normalize the university database to 1NF, 2NF, 3NF and BCNF.
10	To perform transactions while maintaining the ACID properties.
11	To implement high level language extensions with Cursors and Triggers.
12	To implement Procedures and Functions.
13	To design a database for the Banking System.
14	To design a database for the Library Information System.
15	To design a database for the Student Information System.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
HSMC-412	Corporate Etiquette and Personality Development	HSMC	1	0	0	2	50	0	50

Course Outcomes

After completion of this course the student will be able to	
CO1	Appreciate social etiquettes, norms for professional and personal behaviour.
CO2	Use modern tools for making effective presentations.
CO3	Develop competence in professional verbal and non-verbal communication.
CO4	Develop self-awareness and articulate a personal vision.
CO5	Gain insights into emotional intelligence and work effectively in teams.

List of Activities for Corporate Etiquette and Personality Development

S.No.	Activity
1	Polishing Business Manners: Handshake, gifts, visiting cards, humour, office behaviour etc.
2	Power Dressing: Wardrobe Etiquette, Grooming for Success.
3	Body Language, Poise, and Eye Contact.
4	Prepare effective presentations using different modern tools.
5	Understanding the Art of Entertaining: Playing a Gracious Host.
6	Work on Importance of Self Grooming and Personal Hygiene, Gestures and Postures.
7	Language Lab Effective Communication Skills: Word Power enhancement; Attention to Detail: spacing, punctuation, spelling, and other finer aspects; Verbal and non-verbal communication skills; Getting rid of inhibitions and building confidence; Assertive and Submissive communication; Using Language for Convincing and Persuasion; Art of asking Questions.
8	Handling difficult situations with grace, style, and professionalism.
9	To carry out a personal SWOT Analysis.
10	Gender Sensitization.
11	Develop empathy towards customers and colleagues by listening to their problems and understanding them emotionally.
12	Know your colleagues.
13	Apologizing the right way.

*The course would involve interactive sessions, individual and group exercises, role plays, situation-handling, and experience-sharing. Selected video films will complement these.

Course Code	Specialization Track	Course Type	Cd	L	T	P	Marks		
							Mid-Semester	End-Semester	Total
MDC-AI/CS/CC/IoT /DS-401	Artificial Intelligence and Machine Learning/ Cyber Security/ IoT/ Data Science/ Cloud Computing	MOOC	5	5	0	0	25	125	150

The students shall be required to complete a MOOC for duration 8-12 weeks corresponding to the opted specialization track. Such MOOCs shall be approved by the Departmental Academic Committee and updated from time to time.

The Departmental Academic Committee shall assess the student work based on the following criteria:

1. Submission of course completion certificate- **75 Marks**
2. Mid Semester Test - **25 Marks**
3. End- Semester Test- **50 Marks**

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Max. Marks
1	COM-504	Professional Industrial Training/ Internship	Summer Internship	3	0	0	0	100

Note: Students are required to complete the professional industrial training/ Internship during summer semester breaks.

Semester5

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
1	COM-501	Python Programming-I	PCC	4	3	1	0	50	100	150
2	COM-502	Computer Networks	PCC	4	3	1	0	50	100	150
3	COM-503	Microprocessors	PCC	4	3	1	0	50	100	150
4	HSMC-501	Open Elective1	OEC	3	2	1	0	50	100	150
5	COM-511	Python Programming Lab	PCC	2	0	0	4	50	0	50
6	COM-512	Computer Networks Lab	PCC	1	0	0	2	50	0	50
7	COM-513	Microprocessor Lab	PCC	1	0	0	2	50	0	50
8	HSMC-511	Placement Preparation-I	HSMC	2	0	0	4	50	0	50
Total				24*	27					900*

* Note: Summer Internship Credits/Marks are also added in this semester's marksheet.

Hons. / Minor Degree Course

Course Code	Specialization Track	Course Type	Cd	L	T	P	Marks		
							Mid-Semester	End-Semester	Total
MDC-AI/CS/CC/IoT/DS-501	Artificial Intelligence and Machine Learning Cyber Security IoT Data Science Cloud Computing	MOOC	5	5	0	0	25	125	150

*Note: Student can enroll in only one course from chosen Hons. /Minor Degree Course. Total Marks is 150 for any one of the chosen course in the Hons. / Minor Degree Course.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-501	Python Programming-I	PCC	4	3	1	0	50	100	150

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Demonstrate thorough understanding of basic data types in Python and permissible operations on them
CO2	Apply available programming constructs in Python to solve programming problems
CO3	Apply object-oriented decomposition to model real-world scenarios and applications.
CO4	Implement functions and utilize modules in Python to deliver the desired functionality.
CO5	Demonstrate ability to use advanced I/O operations, exception handling and debugging.

Detailed Syllabus

Section-A

Unit 1: Introduction to Python Programming Language: Introduction to Python Language, Strengths and Weaknesses, Built-in Functions, Conversions, Numeric Data Types, String Operators, String Slices, String Operations, String values, Naming Conventions, Integrated Development and Learning Environment, Dynamic Types.

(10 Hrs)

Unit 2: Data Collections and Language Component: Introduction, Control Flow and Syntax, Indenting, If Statement, Relational Operators, Logical Operators, Boolean, Bit Wise Operator, The While Loop, break and continue, For Loop, Lists, Tuples, Sets, Dictionaries, Sorting Dictionaries, Copying Collections.

(8 Hrs)

Unit 3: Object and Classes: Classes in Python, Principles of Object Orientation, Creating Classes, Instance Methods, File Organization, Special Methods, Class Variables, Inheritance, Polymorphism, Type Identification, Custom Exception Classes.

(10 Hrs)

Section-B

Unit 4: Functions and Modules: Introduction, defining your own Function, Parameters, Function Documentation, Keyword and Optional Parameters, Passing Collections to a Function, Variable Number of Arguments, Scope, Functions- “First Class Citizens”, Passing Function as an Argument, Mapping Functions in a Dictionary, Lambda, Modules, Standard Modules- ‘sys’, ‘math’, ‘time’, The ‘dir’ Function.

(12 Hrs)

Unit 5: I/O and Error Handling in Python: Introduction, Data Streams, creating your own Data Streams, Access Modes, Writing Data to a File, Reading Data from a File, Additional File Methods, Using Pipes as Data Streams, Handling I/O Exceptions, Working with Directories, Metadata, Errors, Run Time Errors, The Exception Model, Exception Hierarchy, and Handling Multiple Exceptions.

(12 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Think Python	Allen B. Downey	O'Reilly	2nd (2015)
2	Learning Python	Mark Lutz	O'Reilly	5th (2013)
3	Core Python Programming	Wesley J. Chun	Pearson	2nd (2007)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Dive Into Python 3	Mark Pilgrim	Apress	1st (2009)
2	Python Programming	Mark Lutz	O'Reilly	4th (2010)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-502	Computer Networks	PCC	4	3	1	0	50	100	150

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Articulate the fundamental concepts in computer networks and justify their layered implementation.
CO2	Appreciate the challenges involved and available protocols for facilitating low-level data communication.
CO3	Analyze the design issues in ensuring error free data construction, deconstruction and transmission across devices.
CO4	Justify the application of appropriate routing algorithms in different scenarios based on through Understanding of prevalent routing algorithms.
CO5	Articulate the role of higher network layers in the OSI models and the working of common network Applications.

Detailed Syllabus

Section A

Unit 1: Introduction: Data Communication-communication system, synchronous and asynchronous systems, serial and parallel systems, dataflow-simplex, half-duplex, full-duplex, computer network-uses of computer network, categories of computer networks, protocol and standards, Reference Model-OSI and TCP/IP reference model, their comparison and critique, Network Topologies.

(12 Hrs)

Unit 2: Physical Layer: Data Transmission-Digital to Digital Conversion-Line Coding Scheme, Transmission Media, RS-232 Interface, Switching mechanisms and Comparison –circuit, packet, message, Modem and its types.

(11 Hrs)

Unit 3: Data Link Layer: Design Issues, Error Detection and Correction, Flow Control-Elements of data link protocol, Sliding Window Protocol, Example of Data Link Protocol (HDLC), Channel Allocation Problems, Multiple Access Protocol-ALOHA, Carrier Sense Multiple Access Protocols, Collision Free Protocols, IEEE standards-802.3, 802.4, 802.5.

(9 Hrs)

Section B

Unit 4: Network Layer : Design Issues, Routing Algorithms–the optimality principle, shortest path algorithm, flooding, distance vector routing, link state routing and hierarchical routing, Congestion Control- principles prevention policies, congestion control in virtual circuit subnet and datagram subnets, Traffic shaping algorithm - leaky bucket algorithm, token bucket algorithm, QOS, IP protocol, IP addresses, Internet Multicasting, Introduction to IPV6, IPV4 vs. IPV6, Internetworking devices –concept of Internetworking, Repeaters, Hub, Bridges, Switches, Routers, Gateways.

(8 Hrs)

Unit 5: Transport, Session, Presentation and Application Layer: Transport Layer Services, Primitives, Issues, elements of transport protocol, Introduction to TCP and UDP, Design Issues, FTP, DNS, E-Mail, Introduction to WWW, Firewalls.

(8 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Computer Networks: A System Approach	W. Stallings	Pearson Education	2nd (2010)
2	Computer Networks: A System Approach	L. L. Peterson	Morgan Kauffman	5th (2011)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Computer Networking: A Top-Down Approach Featuring the Internet	Kurose & Ross	Pearson Education	3rd (2005)
2	Computer Networks and Internets with Internet Applications	Coumar & Douglas	Prentice Hall	4th (2004)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-503	Microprocessors	PCC	4	3	1	0	50	100	150

Course Outcomes:

At the end of the course the student will be able to:	
CO 1	Appreciate the architectural elements in a Microprocessor and their functionality.
CO 2	Analyze, design, implement kand test various assembly language programs of moderate complexity.
CO 3	Classify the instruction set of 8085 and 8086 microprocessors and distinguish the use of different instructions and apply them in assembly level programming.
CO 4	Appreciate the evolution of RISC and ARM processors and the improvements over older microprocessor architectures.
CO 5	Appraise the architecture of advanced processor families

Detailed Syllabus

Section-A

Unit 1: Architecture of 8085: Block diagram, Pin Description of 8085, Instruction Set and Instruction Format, Addressing Modes, Looping, Counting and Indexing. 8085 Interrupts, Interrupt handling in 8085, Enabling, disabling and masking of interrupts.

(8 Hrs)

Unit 2: Programmable Interface Devices: - Basics of Programmable I/O, General Purpose Programmable Peripheral Devices – 8255A, 8259A, Direct Memory Access Controller – 8237.

K9 **(12 Hrs)**

Unit 3: Architecture of 8086: Memory Address space and data organization, segment registers and memory segmentation, generating memory addresses, IO address space, addressing modes, Minimum mode and Maximum mode.

(8 Hrs)

Section-B

Unit 4: RISC Processors and ARM: The RISC revolution, RISC Architecture and its characteristics, Pipeline bubbles, accessing external memory in RISC systems, Reducing the branch penalties, Branch prediction, ARM processors, ARM registers, ARM instructions, The ARM built-in shift mechanism, ARM branch instructions, sequence control, Data movement and memory reference instructions.

(12 Hrs)

Unit 5: Advanced Microprocessors: Concept of core processor. Basic features of Advanced Microprocessors: Pentium I3, I5, and I7.

(8 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Microprocessor Architecture, Programming and Applications with 8085	Ramesh S. Gaonkar	Penram International Publishing	2nd (2010)

2	Microprocessor and Interfacing	Douglas V. Hall	McGraw Hill Education	5th (2017)
3	Advanced Microprocessors and Peripherals	K. M. Bhurchandani and A. K. Ray	McGraw Hill Education	3rd (2018)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Introduction to Microprocessors	Aditya Mathur	McGraw Hill Education	3rd (2005)
2	Microprocessor & Interfacing	AP Godse	Technical Publications	4th (2014)
3	The Pentium Microprocessor	James Antonakons	Pearson Education	5th (2020)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
HSMC-501(A)	Business Environment and Ethics	OEC	3	2	1	0	50	100	150

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Articulate the types of business environments and their importance in overall business sustenance and growth
CO2	Apply the process of Environmental Scanning in different settings using different models for analysis
CO3	Appreciate the overall economic environment and analyze different global economic systems and policies
CO4	Examine significant contemporary ethical issues and challenges in business environments and their implications.
CO5	Appreciate different models of corporate social responsibility and the roles and responsibilities of different stakeholders.

Detailed Syllabus

Section-A

Unit 1: The concept of Business Environment, Significance and Nature; Types of Business Environment: Internal and External Environment, micro and macro environment. Impact of business environment on business decision.

(8 Hrs)

Unit 2: Environment Scanning: Meaning, Nature and Scope; The process of Environmental Scanning; PEST Analysis; Competitor's Analysis: Porter's Five Force Model; SWOT Analysis.

(8 Hrs)

Unit 3: Concept and nature of economic environment; Critical elements of economic environment; Basic economic systems: Capitalism, socialism, mixed; New economic policies: Liberalization, privatization and globalization; FEMA; Monetary and fiscal policies.

(8 Hrs)

Section-B

Unit 4: Introduction to Business ethics, Definition, Need, Importance, Nature, Scope, Objectives of Business Ethics, Factors Influencing Business Ethics, Characteristics of Business Ethics, Ethical Aspects in Marketing, Finance, HR, Ethical Decision Making.

(8 Hrs)

Unit 5: Corporate Social Responsibility: Concept, Scope & Relevance and Importance of CSR in Contemporary Society. CSR towards - Share holders, Creditors and Financial Institutions, Government, Consumers, Employees and Society.

(8 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Essentials of Business Environment	Aswathapa, K	Himalaya Publishing House	8th (2011)
2	Business Ethics Text & Cases	C.S.V Murthy	Himalaya Publishing House	8th (2018)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Business Environment	Francis Cherunilam	Himalaya Publishing House	28th (2019)

2	Business Ethics: An Indian Perspective	A.C Fernando, K. P Muralidheeran, E. K Satheesh	Pearson Education	3rd (2019)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
HSMC-501(B)	Organizational Behavior	OEC	3	2	1	0	50	100	150

Course Outcomes:

At the end of the course the student will be able to: -	
CO 1	Illustrate the nature and skills required for managerial work.
CO 2	Appreciate and articulate important Management Functions including Planning, Organizing, Staffing, Directing and Controlling.
CO 3	Analyze individual and group behavior using existing theories and framework to appreciate its implications on organizational behavior.
CO 4	Appreciate group/team dynamics and evaluate the appropriateness of various conflict management strategies used in organizations.
CO 5	Analyze different leadership and motivational theories in affecting organizational change and development.

Detailed Syllabus

Section A

Unit 1: The Management Process and the nature of managerial work: Managerial skills and roles, Development of management thoughts, Contemporary challenges in management, Difference between Management and Administration.

(8 Hrs)

Unit 2: Organization Structure & Culture: Meaning of Organization Design and Structure, Basic elements of Organization Structure, Types of Organization Design, Meaning and Nature of Organization Culture - Types of Culture, Creating and Maintaining Organization Culture, Managing Cultural Diversity.

(8 Hrs)

Unit 3: Organizational Behavior: Personality-Trait Theory, Big Five Models, Perception, Attitudes, Johari Window, Transactional Analysis: Ego states, Life positions, Transactions.

(8 Hrs)

Section B

Unit 4: Group Behavior: Types of Groups, Stages of Group Development, Group conflict and Resolution strategies, Group Cohesiveness & Group Decision Making, Effective Team Building. Work Stress: Concepts, Nature, Causes and Coping Strategies.

(8 Hrs)

Unit 5: Leadership: Approaches and Styles, Gender and Leadership, Leadership effectiveness, Motivation: Types, Importance and Theories. Organizational Change and Development.

(8 Hrs)

Text Books

S. No.	Name of the Suggested Books	Name of Author	Publisher Name	Edition (Pub. Yr.)
1	Organizational Behavior	Stephen P. Robbins	Prentice-Hall of India	11th (2015)

2	Essential of Management	Koontzo Donnel	Mc Graw Hill	11th (2020)

Reference Books

S. No.	Name of the Books	Name of the Author	Publisher Name	Edition (Pub. Yr.)
1	Principles of Management.	Tripathi P.C & Reddy, P. N	Tata Mc Graw Hill	9th (2012)
2	Principles and Practices of Management	L.M.Prasad	Sultan Chand & Sons	20th (2017)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
HSMC-501(C)	Managerial Economics	OEC	3	2	1	0	50	100	150

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Understand and articulate the concepts of managerial economics and its relationship to business operations.
CO2	Apply marginal analysis and demand forecasting under different market conditions.
CO3	Analyze the market structure including causes and consequences of different market conditions
CO4	Apply the process of calculating national income, analyze its components including circular flow of income and income identities
CO5	Analyze different fiscal and monetary policies and the implications

Detailed Syllabus

Section-A

Unit 1: Characteristics of Managerial Economics – Nature and Scope of Managerial Economics – Importance of Managerial Economics – Basic Economic Tools in Managerial Economics – Managerial Economist's Role and Responsibility.

(6 Hrs)

Unit 2: Demand Estimations for Major Consumer Durables and Non-durable Products – Demand Forecasting Techniques: Statistical and Non-statistical Techniques.

(7 Hrs)

Unit 3: Definition of Market – Market Structure (Perfect Competition, Imperfect Competition) – Price Determination – Firm's Equilibrium in Perfect Competition, Monopoly, Monopolistic, Oligopoly and Duopoly.

(7 Hrs)

Section-B

Unit 4: National Income – Concepts – Methods – Measurement of National Income – GDP and GVA – Business Cycles – Nature – Phases – Causes – Inflation – Causes and Control – Deflation and Stagflation.

(7 Hrs)

Unit 5: Fiscal Policy – Deficits – Budgetary Deficit – Primary Deficit – Revenue Deficit – Fiscal Deficit – Objectives of FRBM Act – Monetary Policy – Objectives – Repo Rate – Reverse Repo Rate – CRR – SLR – Finance Commission – Role and Objectives.

(8 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Managerial Economics	Maheshwari	PHI Learning	3rd (2013)
2	Managerial Economics	Peterson, Lewis Jain	Pearson	4th (2009)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Managerial Economics	Amit Ahuja	Vikas Publishing House	9th (2020)
2	Managerial Economics	DN Dwivedi	Vikas Publishing House	8th (2015)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
HSMC-501(D)	Introduction to Human Psychology	OEC	3	2	1	0	50	100	150

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Illustrate the various perspectives and theories in the field of modern psychology
CO2	Understand the perspectives of learning theories and motivation
CO3	Articulate the applicability of psychology in ensuring overall human well-being
CO4	Appreciate the complexities in the decision-making process and change management in organizations
CO5	Demonstrate understanding of organizational culture, strategies for culture building and relevant issues.

Detailed Syllabus

Section-A

Unit 1: Introduction, Definition and Scope of Psychology; Psychology as a science, methods of psychology; Subfields of psychology; Psychology in modern India.

(6 Hrs)

Unit 2: Principles and applications of Classical conditioning, Operant conditioning, and Observational learning; Cognitive influences on learning; Perspectives on motivation.

(8 Hrs)

Unit 3: Application of Psychology: Stress-management, Burnout: Causes, effect and copingwell-being; Self-development: Application of Psychology in building memory and creativity.

(5 Hrs)

Section-B

Unit 4: Decision making process, individual influences, group decision process; Forces for change in Organization, Resistance to change, Lewin's Change Model.

(8 Hrs)

Unit 5: Characteristics of the Workplace: Organizational Culture and climate; Strategies for reinforcement of culture, Functions of organizational culture, Organizational Socialization, Assessing Cultural Values and Fit, Cross-Cultural issues

(8 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Industrial Psychology	H. L. Kalia	The Associated Publishers	1st (2006)
2	Industrial and Organizational Psychology	Cary, L. Cooper and Edwin, A. Locke,	Blackwell Publishing	1st (2000)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Industrial and Organizational Psychology	Spector	Pearson Education, Inc.	6th (2010)
2	Organizational Behavior	Robbins, Stephen	Prentice Hall, India	11th (2012)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-511	Python Programming Lab	PCC	2	0	0	4	50	0	50

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Demonstrate thorough understanding of basic data types in Python and permissible operations on them
CO2	Apply available programming constructs in Python to solve programming problems
CO3	Apply object-oriented decomposition to model real-world scenarios and applications.
CO4	Implement functions and utilize modules in Python to deliver the desired functionality.
CO5	Demonstrate ability to use advanced I/O operations, exception handling and debugging.

List of Activities for Python Programming Lab

S. No	Activities
1.	Write a program to demonstrate type checking of various data types and demonstrate the use of following built in functions in python: abs(), len(), min(), round(), isalnum(), type().
2.	Write a program to illustrate iteration over the list and dictionary.
3.	Consider the string “Welcome to Python world”. Perform the following operations: Count the number of alphabets in the given string. To extract characters in the given, range from the given string. Check if the string is alphanumeric or not.
4.	Write a program to perform searching activity using Linear and binary search.
5.	Write a program to reverse every kth row in a matrix.
6.	Write a program to create a list of student’s records and search a student record using a dictionary.
7.	Create a database using lists and tuples.
8.	Write a python program to convert temperature to and from Celsius to Fahrenheit.
9.	Write a program to implement an employee management system using classes, instances and inheritance.
10.	Write a program to create a Simple Calculator using a switch case and function for every operation.
11.	Write a python program that accepts the length of three sides of a triangle as inputs. The program should indicate whether or not the triangle is a right-angled triangle. (Use Pythagorean theorem) Also find out its area using Heron’s formula.

12.	Create a function max_of_three that takes three numbers as arguments and returns the largest of them and also create a parameter function that checks whether a given number is Armstrong or not.
13.	Write a program that inputs a text file. The program should print all of the unique words in the file in alphabetical order.
14.	Create a text file “MyFile.txt” in python and ask the user to write separate 3 lines with three input statements from the user.
15.	Create a python program to implement Morse Code Translator.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-512	Computer Networks Lab	PCC	1	0	0	2	50	0	50

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Apply the low-level computer networking concepts via standard hardware and software-based simulation tools so as to practically implement the understanding of standardized models.
CO2	Devise network topologies and emulate modern computer networks using Cisco packet tracer.
CO3	Evaluate the effectiveness of Link state and Distance Vector routing protocols in different real-world networking scenarios
CO4	Analyze different network configuration modes so as to effectively develop the best network practices.
CO5	Create basic client-server frameworks through the implementation of primitives of Socket programming.

List of Activities for Computer Networks Lab

S. No.	Activities
1	Fabrication of cables: Making Straight, Rollover and Cross-Over cables using RJ-45 connector and crimping tool.
2	Familiarization with Network Devices in Detail.
3	Configure a Router with Different VLANS and analyze the security of Individual VLANS.
4	Basic LAN Setup via Packet Tracer.
5	Familiarization with basic Network Commands along with different configuration modes.
6	Configure a Network Topology using Packet Tracer.
7	Familiarization with different Subnetting Techniques.
8	Simulate Static Routing in a topology.
9	Simulate RIP protocol in a topology.
10	Simulate OSPF protocol in a topology.
11	Simulate EIGRP protocol in a topology.
12	Using TCP/IP sockets, write a client – server program to make the client send the file name and to make the server send back the contents of the requested file if present.

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Computer Networks: A System Approach	W. Stallings	Pearson Education	2nd (2010)
2	Computer Networks: A System Approach	L.L. Peterson	Morgan Kauffman	5th (2011)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Computer Networking: A Top-Down Approach Featuring the Internet	Kurose & Ross	Pearson Education	3rd (2005)
2	Computer Networks and Internets with Internet Applications	Coumar & Douglas	Prentice Hall	4th (2004)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-513	Microprocessor Lab	PCC	1	0	0	2	50	0	50

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Apply programming strategies and select proper mnemonics to run the program on the training boards.
CO2	Practice different types of programming techniques keeping in mind technical issues and evaluate possible causes of discrepancy in practical experimental observations in comparison.
CO3	Develop testing and experimental procedures on microprocessor and analyze their operation under different cases.
CO4	Analyze, design, test and implement various assembly language programs of moderate complexity.
CO5	Design/modify the software to optimize its run-time behavior.

List of Activities for Microprocessor Lab

S. No	Activities
1.	Block Transfer - Data bytes are stored in memory locations from XX50H to XX5FH To insert an additional five bytes of data, it is necessary to shift the data string by five memory location. Write a program to store a data string from XX55H. To XX64H. Use any 16 bytes of data to verify your program.
2.	Addition with Carry: Six bytes of data are stored in memory locations starting at XX50H. Add all the data bytes. Use register B to save any carry generated while adding the data bytes. Store the sum at two consecutive memory locations XX70H and XX71H.
3.	Checking for a particular data byte: A set of eight readings is stored in memory location starting at XX50H. Write a program to check whether a byte 40H exists in the set. If it does, stop checking, and display its memory location, otherwise output FFH.
4.	Write a program to generate square wave of 1 KHZ frequency using 8085 microprocessor kit.
5.	Write a program to generate saw tooth wave of 1 KHZ frequency using 8085 microprocessor kit.
6.	Write a program to generate triangular wave of 1 KHZ frequency using 8085 microprocessor kit
7.	Write an assembly language for division two 8-bit numbers by using microcontroller 8051 kit.
8.	Write an assembly language for multiplying two 8-bit numbers by using microcontroller 8051 kit.
9.	Display name on the LCD using microcontroller 8051.
10.	Generate beep sound on a buzzer using microcontroller 8051.
11.	Study of 8255 interfacing card.
12..	Study of 8257 DMA controller interface card.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
HSMC-511	Placement Preparation-I	HSMC	2	0	0	4	50	0	50

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Comprehend and assimilate subject-specific knowledge as per industry requirements.
CO2	Identify the techniques to approach effective problem-solving.
CO3	Understand the Higher Education ecosystem.
CO4	Learn to communicate effectively with a wide variety of audiences.
CO5	Understand their personality type and interview requisites.

List of Activities for Placement Preparation

S. No	Activities
1.	Practice Computer Fundamentals Introduction to DBMS, SQL Basics, Joins, Normalization, Subquery, Searching and Sorting, Transactions, NoSQL Databases
2.	Programming Basics Introduction to Python, Algorithms, Variables, Datatypes & Operators in Python, Control Structure, Tuple, Unit Testing, Collection, Function, OOPS Concept, Introduction to Java, Polymorphism, OOP Basics, Arrays and String, Abstract, final and Interface, Recursion.
3.	Pseudo Code and Data Structure Code snippet on C, C++, and Java Understanding of programming logic (tested through Pseudo Code Questions Queue, Array, Stacks, Tree, Graphs, Time complexity, Hashing Code Debugging Automata fix
4	Game-Based Aptitude Grid Challenge, Deductive-logical Thinking, Inductive-logical Reasoning, Motion Challenge, Switch Challenge, Digit Challenge
5.	Verbal Ability Reading Comprehension, Sentence Correction, Spotting Error, Speech, and Tenses, Sentence Selection, Sentence Arrangement, word completion
6.	Industry Ready - Foundation Resume Building, Group Discussion, Personal Interviews, Big 5 Personality Tests, LinkedIn Profile Building. Domain Specific Industry Oriented Certification
7.	Participation in Branch specific technical assessments, Coding Competitions, and Hackathons.

8.	Higher Education Explore higher education opportunities in India and abroad, the Scholarship process for international universities, and the competitive entrance examination process.

Course Code	Specialization Track	Course Type	Cd	L	T	P	Marks		
							Mid-Semester	End-Semester	Total
MDC-AI/CS/CC/IoT/DS-501	Artificial Intelligence and Machine Learning Cyber Security IoT Data Science Cloud Computing	MOOC	5	5	0	0	25	125	150

The students shall be required to complete a MOOC for duration 8-12 weeks corresponding to the opted specialization track. Such MOOCs shall be approved by the Departmental Academic Committee and updated from time to time.

The Departmental Academic Committee shall assess the student work based on the following criteria:

1. Submission of course completion certificate- **75 Marks**
2. Mid Semester Test - **25 Marks**
3. End- Semester Test- **50 Marks**

Semester 6

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
1	COM-601	Artificial Intelligence with Computer Vision	PCC	4	3	1	0	50	100	150
2	COM-602	Fundamentals of Blockchain Technology	PCC	4	3	1	0	50	100	150
3	COM-603	Software Engineering	PCC	4	3	1	0	50	100	150
4	COM-604	Theory of Computation	PCC	4	3	1	0	50	100	150
5	COM-611	Artificial Intelligence with Computer Vision Lab	PCC	2	0	0	4	50	0	50
6	COM-612	Mini Project Lab	PCC	1	0	0	2	50	0	50
7	HSMC-611	Placement Preparation-II	HSMC	2	0	0	4	50	0	50
8	NCC-601	MOOC	NCC	0	2	0	0	-	-	S/NS*
Total				21	29					750

Note: S=Satisfactory, NS=Not Satisfactory.

Hons. / Minor Degree Course

Course Code	Specialization Track	Course Type	Cd	L	T	P	Marks		
							Mid-Semester	End-Semester	Total
MDC-AI/CS/CC/IoT /DS-601	Artificial Intelligence and Machine Learning/ Cyber Security/ IoT/ Data Science/ Cloud Computing	MOOC	5	5	0	0	25	125	150

*Note: Student can enroll in only one course from chosen Hons. / Minor Degree Course. Total Marks is 150 for any one of the chosen course in the Hons. / Minor Degree Course.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-601	Artificial Intelligence with Computer Vision	PCC	4	3	1	0	50	100	150

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Design and implement intelligent solutions to classification, regression and clustering problems.
CO2	Design and implement distinguishing features in pattern and object recognition tasks.
CO3	Understand how images are represented numerically and to Implement image processing techniques like Colour and geometric transforms.
CO4	Design to programmatically track a single point over time and understand motion models that define Object Movement over time.
CO5	Implement a method for tracking a set of unique features over time to match features from one image Frame to another.

Detailed Syllabus

Section A

UNIT 1: Introduction to Artificial Intelligence: Basic of AI, history of AI, various techniques of Artificial Intelligence (Machine learning, NLP, Automation & Robotics and Computer vision, Types of AI, Applications of Artificial Intelligence, two ways of Artificial Intelligence work i.e., symbolic and Data Based.

(8 hrs)

UNIT 2: Foundation of Machine learning: Terminology used in ML, ML Process (Building the Model), Supervised ML, Unsupervised ML and Reinforcement ML and its Algorithms, types of problems solved using ML (Regression, Classification, Clustering), Exploratory Data Analysis on Real world Problem i.e., possibility of Rain including the variable dependency.

(8 hrs)

UNIT 3: Image Processing, Computer Vision and Computer Graphics: Computer Vision - Low-level, Mid-level, High-level, Overview of Diverse Computer Vision Applications, Light-Models, Image Acquisition, and Backgrounds: Adaptive Background Subtraction; Camera Geometry, Calibration, and Stereo Interest Operators and Feature Extraction.

(8 hrs)

Section B

UNIT 4: Tracking: Feature Tracking (Selecting features, Nose Tracking); Direct Method (Image-Based Tracking): Head Tracking, Face feature tracking; Model-based Tracking: (Hand tracking, Body tracking, Fingertip Tracking), Kalman Filtering, Particle Filtering and Mean shift tracking.

(8 hrs)

UNIT 5: Recognition: Space-Time Methods: Discriminant Analysis, Bayesian Network Classifiers; Temporal Methods; Model-Based Methods (Markov-Dynamic Networks); Hidden Markov Models.

(8 hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Artificial Intelligence: A Modern Approach	Stuart Russell and Peter Norvig	Pearson	4th (2020)
2	Computer Vision - A modern Approach	D. Forsyth and J. Ponce	McGraw-Hill	2nd (2018)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Artificial Intelligence: Strategies and Techniques for complex problems solving	George Luger	Addison Wesley	6th (2008)
2	Introductory Techniques for 3D Computer Vision	E. Trucco and A. Verri	Prentice Hal	2nd (2009)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-602	Fundamentals of Blockchain Technology	PCC	4	3	1	0	50	100	150

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Articulate the history, types and applications of Blockchain.
CO2	Explain the Blockchain architecture in context of different crypto currency.
CO3	Comprehend different consensus algorithms and justify their appropriateness for different applications.
CO4	Gain appreciation of different decentralized applications over Blockchain frameworks.
CO5	Formulate solutions using Blockchain technology for real world applications.

Detailed Syllabus

Section-A

Unit 1: Introduction to Blockchain: Introduction to Block chain – History, Definition, Distributed Ledger, Blockchain Categories – Public, Private, Consortium, Blockchain Network and Nodes, Peer-to-Peer Network, Mining Mechanism, Generic elements of Blockchain, Features of Blockchain and Types of Blockchain, Uses of Blockchain.

(8 Hrs)

Unit 2: Blockchain Architecture: Abstract Models for Blockchain - GARAY model - RLA Model. Multichain: Objective of Multichain, Features, Operation of Bitcoin Blockchain, Blockchain Architecture – Block, Hash, Distributed P2P, Fundamentals of Bitcoin, Ethereum and other popular crypto currencies.

(8 Hrs)

Unit 3: Consensus: Importance of Consensus mechanism, Proof of Work (PoW), Proof of Stake (PoS), Byzantine Fault Tolerance (BFT), Proof of Authority (PoA) and Proof of Elapsed Time (PoET). Scalability aspects of Blockchain consensus protocols. **(8 Hrs)**

Section-B

Unit 4: Blockchain Decentralized Applications: Characteristics of Decentralized application, De-Fi (Decentralized Finance) Applications and use-cases, Smart contracts and decentralized oracle networks (DONs).

(8 Hrs)

Unit 5: Block Chain- Case Studies: Case studies on applications of Blockchain technology in real world applications **(8 Hrs)**

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Blockchain Technology From Theory to Practice	Sudeep Tanwar	Springer	1st (2022)
2	Blockchain Technology: Concepts and Applications	Kumar Saurabh, Ashutosh Saxena	Wiley	1st (2020)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Bitcoin and Cryptocurrency Technologies	Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfede	Princeton University Press	1st (2016)

2	Mastering Bitcoin: Unlocking Digital Cryptocurrencies	Andreas M. Antonopoulos	O'Reilly Media Inc	1st (2015)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-603	Software Engineering	PCC	3	2	1	0	50	100	150

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Analyze software development process models, including agile models and traditional models.
CO2	Demonstrate the use of software life cycle through requirements gathering, choice of process model and design model.
CO3	Apply and use various UML models for software analysis, design and testing.
CO4	Acquire knowledge about the concepts of application of formal specification, case tools and configuration management for software development.
CO5	Analysis of software estimation techniques for creating project baselines.

Detailed Syllabus

Section-A

Unit 1: Software Engineering and Processes: - Introduction to Software Engineering, Software Evolution, Software Characteristics, Legacy Software, Software Crisis: myths and Causes, Software Engineering: A layered Technology, Process Framework, and Software process models (Waterfall, Incremental, and Evolutionary process models and Agile).

(8 hrs)

Unit 2: Requirements Engineering: Problem Analysis, Requirement elicitation and Validation, Requirements modeling: Scenarios, Information and analysis classes, flow and behavioral modeling, documenting Software Requirement Specification (SRS).

(5 hrs)

Unit 3: Software Planning and Project management: -Software project management Process: Software scope, Resources, Software Metrics, Software project estimation, Decomposition techniques, Empirical estimation model: COCOMO, Software project scheduling, Risk Analysis, Software acquisition.

(6 hrs)

Section-B

Unit 4: Software Design and construction: System design principles: levels of abstraction (architectural and detailed design),coupling and cohesion, Structured design (top-down functional decomposition), object-oriented design, event driven design, component-level design, test driven design, aspect oriented design , function oriented, service oriented, Design patterns, Coding Practices: Techniques, Refactoring, Integration Strategies, Internal Documentation, Data Flow Diagrams transform analysis, transaction analysis, Transform and Transaction Mapping.

(8 hrs)

Unit 5: Software Quality Assurance, Testing and Maintenance: -Software quality and software quality assurance, Formal Technical Reviews, Software quality metrics: McCall's Quality Factors, Software Reliability, Software Testing Fundamentals, White box Testing, Basic Path Testing, Control Structure Testing, Black Box Testing. Software Testing Strategies, Unit Testing, Integration Testing, Validation Testing, System Testing, Maintenance characteristics, Reverse Engineering, Re-engineering.

(12 hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Software Engineering, A	Pressman S. R. and	McGraw Hill	9th (2020)

	Practitioner's Approach	Maxim R. B.	International	
2	Software Engineering	Sommerville I	Addison-Wesley Publishing Company	10th (2017)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Software Engineering: A Methodical Approach	Foster C. E	Apress	2nd (2021)
2	The Unified Modeling Language User Guide	Booch G., Rambaugh J., Jacobson I	Pearson	2nd (2005)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-604	Theory of Computation	PCC	4	3	1	0	50	100	150

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Understand the key notions of computation, such as algorithm, computability, decidability, reducibility, and complexity, through problem solving.
CO2	Understand the models of computation, including formal languages, grammars and automata, and their connections.
CO3	Articulate and explain the Church-Turing thesis and its significance.
CO4	Analyze and design finite automata, pushdown automata, Turing machines, formal languages, and grammars.
CO5	Solve computational problems regarding their computability and complexity and prove the basic results of the theory of computation.

Detailed Syllabus

Section-A

Unit 1: Finite Automata: Deterministic Finite Automata (DFA), Designing, Non- deterministic finite Automata (NFA) without E-moves, Conversions, Equivalence, NFA with E-moves, Regular expression designing, Finite machine with output assigned, Moore and mealy machines, Conversion and Equivalence, Myhill-Nerode Theorem.

(12 Hrs)

Unit 2: Regular Grammar & Context free Languages: Context free Grammar, Context free Languages, reduced form of Grammar, Ambiguous and Non- Ambiguous grammar, acceptors and generators, Relations between Classes of Languages, Pumping lemma of regular sets, Chomsky's hierarchy of languages, derivation Trees, CYK Algorithm for CFL Membership, Testing emptiness of CFLs.

(10 Hrs)

Section-B

Unit 3: Turing Machines: Church Testing Hypothesis, Turing Computability, Non- deterministic, Multitape and other versions of Turing machines, Churches Hypothesis, Primitive Recursive functions, Universal Turing machines, decidability, Halting problem, Stack Automata.

(8 Hrs)

Unit 4: Push Down Automata: Definition, Model, Acceptance of CFL, Acceptance by Final State and Acceptance by Empty stack and its Equivalence, Equivalence of CFG and PDA.

(6 Hrs)

Unit 5: Recursive and Recursively Enumerable Languages (rel): Properties of recursive and recursively enumerable languages, Context sensitive language and linear bounded automata (LBA), Chomsky hierarchy, Decidability, Post's correspondence problem (PCKP), undecidability of PCP.

(6 Hrs)

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Introduction to Automata Theory, Languages and Computation	J Hopcroft, JD Ullman, R Motwani,	Pearson	3rd (2008)
2	Elements of the Theory of Computation	Lewis, H.R. and	Pearson	2nd (2015)

		Papadimitriou		

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Introduction to Automata Theory, Languages and Computation	John E. Hopcroft and Jeffrey D. Ullman.	Pearson	3rd (2008)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-611	Artificial Intelligence and Machine Learning Lab	PCC	2	0	0	4	50	0	50

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Design and implement intelligent solutions to classification, regression and clustering problems using Python.
CO2	Design and implement distinguishing features in pattern and Image Processing tasks.
CO3	Illustrate and apply clustering algorithms and identify its applicability in real life problems.
CO4	Apply learning approaches and acquire skills to design, analyze and develop algorithms and implement them using high level programming languages
CO5	Design and implement solutions to classification problems using CNN.

List of Activities for Artificial Intelligence and Machine Learning Lab

S. No	Activities
1.	Implement and demonstrate the Data Processing for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file.
2.	For a given set of training data examples stored in a .CSV file, implement and demonstrate the Data Visualization to output a description of the set of all hypotheses consistent with the training examples.
3.	Assuming a set of data that need to be specifically used in the dataset applies the Data Wrangling by reading the training data from a .csv file.
4.	Implement the data Processing, Data visualization and Data wrangling on any real-world problem i.e., Covid _19 dataset to view the active cases on the world Map using Chloropleth and Also Plot the cases.
5.	Implement the Linear regression on the salary Prediction based on a given set of training data samples. Read the training data from a .CSV file.
6.	Implement the KNN IRIS Image Classification based on a given set of training data samples. Read the training data from a .CSV file.
7.	Implement the web scraping on Amazon website or any shopping site by importing the requests and the Beautiful Soup.
8.	Implement the Image processing basics: Converting Image to RGB to BGR and to Grayscale, create a White / Black Color Image, how to draw shapes on image.
9.	Implement the detection of Eye or face or Smile from the Image.
10.	Apply the sentimental Analysis using Natural language Processing on Live Twitter data.

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Artificial Intelligence: A Modern Approach	Stuart Russell and Peter Norvig	Pearson	4th (2020)
2	Computer Vision - A modern approach	D. Forsyth and J. Ponce	McGraw-Hill	2nd (2018)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Artificial Intelligence: Strategies and techniques for complex problems solving	George Luger	Addison Wesley	6th (2008)
2	Introductory Techniques for 3D Computer Vision	E. Trucco and A. Verri	Prentice Hal	2nd (2009)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-612	Mini Project Lab	PCC	2	0	0	4	50	0	50

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Develop a mini project in an emerging domain
CO2	Demonstrate effective use of the appropriate technology stack in the emerging domain.
CO3	Demonstrate competence in using modern tools associated with the emerging domain
CO4	Solve real-world problems using the technology related to the emerging domain
CO5	Develop ability to handle complexity by designing effective constructs

List of Activities for Mini Project Lab

S. No.	Activities
1.	Requirements Gathering: Freeze the functional requirements and design considerations.
2.	Project Setup: IDE and development environment, use of Git, code management and version control, project build and deployment
3.	Project Deployment: Use AWS Cloud to host the project and use advanced features such as auto-scale, backup etc..
4.	Incremental Development: Demonstrate weekly progress during the laboratory during continuous assessment.
5.	Technology Leverage: Demonstrate use of well-known APIs, tools, framework such as APIs and services from Google, Postman, AWS, Azure and other Advanced Python Libraries
6.	Elegance and Quality: Focus on writing high-quality code following industry-standard code conventions, comments and documentation
7.	Functionality Delivery: Test and verify that all functional requirements and design considerations are fully met.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
HSMC-611	Placement Preparation-II	HSMC	2	0	0	4	50	0	50

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Apply programming constructs to solve puzzles/problems from a placement perspective
CO2	Develop problem solving skills on online coding platforms
CO3	Demonstrate logical reasoning and aptitude in online tests
CO4	Solve complex problems in data structures and algorithms
CO5	Demonstrate confidence and poise in personal interviews

List of Activities for Placement Preparation

S. No	Activities
1.	Advanced Reasoning: Data Sufficiency, Syllogism, Number Series, Word pattern, Blood relations, Coding and Decoding, Distance and Direction, Visual Reasoning, Logical Venn Diagram based DI, Paper Cuts and folds, Seating Arrangement
2.	Advanced Quantitative Aptitude: Logarithms, Permutation and Combinations, Probability, Pipes and Cisterns Geometry, Coordinate Geometry, Clocks & Calendar Time Speed Distance, LCM & HCF, Series and Progressions, Sudoko, Mensuration, Data Interpretation, Simplification and Calculation
3.	Advanced Coding Problem Solving Sessions: C, C++, Python, and Java
4.	Competitive Coding and Data Structure Algorithms Problem Solving Sessions
5.	Industry Ready – Advanced: Professional networking through LinkedIn, Mock Interviews with External Experts, Group Discussions, Understanding and participation in Campus Hiring Process, Industry certification on Cloud Foundation and Computer Vision.
6.	Participation in a national level employability aptitude test to assess the learner's ability on quantitative, verbal, logical reasoning, soft skills, and technical skills.
7.	Participation in Branch specific technical assessments, Coding Competitions, and Hackathons.
8.	Career Counseling and Career Progression.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
NCC-601	MOOC	NCC	0	2	0	0	-	-	S/NS*

*Note: S=Satisfactory, NS=Not Satisfactory

As part of this elective, student will undertake MOOC's pre-approved by the department in the following suggested areas:

A. Quantum Computing
B. Metaverse
C. Advanced Python
D. Full-Stack Development
E. UI/UX Development

Course Outcomes:
Defined by the online platform for the specific course chosen

Course Code	Specialization Track	Course Type	Cd	L	T	P	Marks		
							Mid-Semester	End-Semester	Total
MDC-AI/CS/CC/IoT /DS-601	Artificial Intelligence and Machine Learning/ Cyber Security/ IoT/ Data Science/ Cloud Computing	MOOC	5	5	0	0	25	125	150

The students shall be required to complete a MOOC for duration 8-12 weeks corresponding to the opted specialization track. Such MOOCs shall be approved by the Departmental Academic Committee and updated from time to time.

The Departmental Academic Committee shall assess the student work based on the following criteria:

1. Submission of course completion certificate- **75 Marks**
2. Mid Semester Test - **25 Marks**
3. End- Semester Test- **50 Marks**

Semester 7

Scheme-1

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
1	COM-704	Professional Industrial Training/ Internship	Summer Internship	4	0	0	0	-	150	150

Internship Outcomes:

After completion of the internship the student will be able to:	
CO1	Identify a relevant problem or functional area in consultation with industry expert or designated mentor and the expected outcomes.
CO2	Perform detailed analysis of the problem in line with expected outcomes.
CO3	Undertake the project work/field work to effectively attain the expected outcomes.
CO4	Demonstrate proficiency in professional communication, teamwork, and work ethics.
CO5	Prepare a formal report outlining the work done, outcomes and demonstrate deep understanding of the work done.

Students are required to undertake 4 to 6 weeks of Practical Training after the final examinations of 6th semester in the field of Computer Science and Engineering in a well-established and recognized organization. The Training and Placement Cell shall accord final approval to the internship request made by the student.

Guidelines for evaluation of Industrial Training:

The evaluation shall be done by the departmental committee by the end of 7th semester. The committee shall have a convener and at least two members. The distribution of marks is given below:

S. No.	Evaluation Component	Marks
1	Presentation (50%)	75
2	Report (50%)	75
Total Marks		150

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-701(A)	Cloud Computing and Services	PEC	3	2	1	0	50	100	150

Course Outcomes:

At the end of the course the students will be able to:	
CO1	Articulate the main concepts, key technologies, strengths, limitations of cloud computing and the possible applications for state-of-the-art cloud computing.
CO2	Identify the architecture and infrastructure of cloud computing, including cloud delivery and deployment models.
CO3	Analyze the performance, scalability, and availability of the underlying cloud technologies and software.
CO4	Develop the core issues of cloud computing such as security, privacy, and interoperability.
CO5	Evaluate the appropriate cloud computing solutions and recommendations according to the applications used.

Detailed Syllabus

Section-A

Unit 1: Cloud Computing Overview: Origins of Cloud computing – Cloud components - Essential characteristics – On-demand self-service, Broad network access, Location independent resource pooling, Rapid elasticity, Measured service, Comparing cloud providers with traditional IT service providers, Roots of cloud computing. **(06 Hrs.)**

Unit 2: Cloud Insights: Architectural influences – High-performance computing, Utility and Enterprise grid computing, Cloud scenarios – Benefits: scalability, simplicity, vendors, security, Limitations – Sensitive information - Application development- security level of third party - security benefits, Regularity issues: Government policies.

(08 Hrs.)

Unit 3: Cloud Architecture-Layers and Models: Layers in cloud architecture, Software as a Service (SaaS), features of SaaS and benefits, Platform as a Service (PaaS), features of PaaS and benefits, Infrastructure as a Service (IaaS), features of IaaS and benefits, Service providers, challenges, and risks in cloud adoption. Cloud deployment model: Public clouds – Private clouds – Community clouds - Hybrid clouds - Advantages of Cloud computing.

(08 Hrs.)

Section-B

Unit 4: Monitoring and Management: Architecture for Federated Cloud Computing, SLA management in cloud computing, performance prediction for HPC on cloud, Data Security in cloud. Legal issues in cloud computing.

(08 Hrs.)

Unit 5: Applications: Best practices in architecting cloud applications in the AWS cloud, building Content Delivery Networks (CDN) using clouds, Resource Cloud Mashups. **(10 Hrs.)**

Text Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Cloud Computing: Principles and Paradigms	Rajkumar Buyya, James Broberg, Andrzej Goscinski	Wiley	1st (2013)

2	Cloud Computing: A Practical Approach	Anthony T.Velte , Toby J. Velte, Robert Elsenpeter	Tata McGraw- Hill	1st (2009)
3	Cloud Security: A comprehensive Guide to secure Cloud Computing	Ronald L. Krutz, Russell Dean Vines	Wiley	1st (2010)

Rference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Cloud computing for Dummies	Judith Hurwitz, Robin Bloor, Marcia Kaufman, Fern Halper	Wiley	2nd (2010)
2	Handbook of Cloud Computing	Borko Furht, Armando Escalante	Springer	1st (2010)

Course Code	Course Name	Course Type	C d	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-701(B)	High Performance Computing	PEC	3	2	1	0	50	100	150

Course Outcomes:

At the end of the course the students will be able to:	
CO1	Explain the basic concepts related to HPC architecture and parallel computing.
CO2	Describe the architecture of computing technology.
CO3	Analyze the design and functioning of existing computing technology architectures.
CO4	Measure the performance of modern parallel computing systems.
CO5	Design the emerging trends in computing technology and analyze the performance of HPC applications.

Detailed Syllabus

Section A

Unit 1: Parallel and Distributed Programming Models: Introduction to high performance computing, basic definitions: cluster, grid, meta-computing, middleware etc., examples of representative applications. Programming models: shared memory, message passing, peer-to-peer. Development of parallel and distributed applications, Design phases, Common parallel patterns, Performance metrics and profiling. **(10 Hrs.)**

Unit 2: Overview of Cluster Computing: The Role of Clusters, Definition and Taxonomy, Distributed Computing, Limitations, Architecture of cluster-based systems, Design Decisions, Network Hardware, Network Software, Protocols Distributed File Systems, Virtualization technologies, Issues in cluster design: performance, single-system-image, fault tolerance, manageability, programmability, load balancing, security, storage. **(8 Hrs.)**

Unit 3: Introduction of Grid Computing: Introduction, Evolution of the Grid, Definitions of Grid Computing, Infrastructure of hardware and software, Grid models, Applications, Examples of usage, Research possibilities / scope in Grid Computing, HPC and Grids, Scheduling HPC applications in Grids, Grid Monitoring Architecture (GMA) – An Overview of Grid Monitoring Systems. **(10 Hrs.)**

Section B

Unit 4: Integrating task parallelism with data parallelism: Introduction and motivation, A model for integrating task parallelism into data parallel programming platforms, Integration of the model into ARC, Design and implementation applications, performance analysis, guidelines for composing user programs, related work. **(10 Hrs.)**

Unit 5: Anonymous remote computing and communication model: Introduction, Location in dependent inter task communication with DP, DP model of iterative grid computations, Design and implementation of distributed pipes. **(9 Hrs.)**

Textbooks

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Grid Computing: A Research Monograph	D. Janakiram	Tata McGraw Hill	1st (2005)
2	High Performance Cluster Computing, Volume 1, Architecture and Systems,	Rajkumar Buyya	Pearson Education	1st (1999)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)

1	Grid Computing	Joshy Joseph & Craig Fellenstein	Pearson Education	1st (2004)
2	Grid Computing: A Practical Guide to technology and Applications	Ahmar Abbas	Charles River Media	1st (2004)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-701(C)	Soft Computing	PEC	3	2	1	0	50	100	150

Course Outcomes:

At the end of the course the students will be able to:	
CO1	Identify intelligent systems leveraging the paradigm of soft computing techniques.
CO2	Analyze the various neural network architectures.
CO3	Design the methodology to solve optimization problems using fuzzy logic.
CO4	Develop the knowledge of genetic algorithm concepts and their applications.
CO5	Evaluate the hybrid system to revise the principles of soft computing in various applications

Detailed Syllabus

Section A

Unit 1: Introduction: What is Soft Computing? Difference between Hard and Soft computing, Requirement of Soft computing, Major Areas of Soft Computing, Applications of Soft Computing. (8 Hrs.)

Unit 2: Artificial Neural Networks: - Basic concepts: Single Layer Perceptron, Multilayer Perceptron, Supervised and unsupervised learning, Back propagation, Networks-Kohen's self-organizing networks, Hopfield network, Feed forward network, Hopfield network. Neural Networks: Neural network models, layers in neural networks and their connections. Instar, Outstar, Weights on connections, Threshold function, Application: Adaline and Madaline. **(6 Hrs.)**

Unit 3: Fuzzy Systems: Fuzzy Set theory, Fuzzy versus Crisp set, Fuzzy Relation, Fuzzification, Minmax Composition, Defuzzification Method, Fuzzy Logic, Fuzzy Rule based systems, Predicate logic, Fuzzy Decision Making, Fuzzy Control Systems, Fuzzy Classification. **(12 Hrs.)**

Section B

Unit 4: Genetic Algorithm: History of Genetic Algorithms (GA), Working Principle, Various Encoding methods, Fitness function, GA Operators- Reproduction, Crossover, Mutation, Convergence of GA, Bit wise operation in GA, Multi-level Optimization. **(8 Hrs.**)

Unit 5: Hybrid Systems: Sequential Hybrid Systems, Auxiliary Hybrid Systems, Embedded Hybrid Systems, Neuro-Fuzzy Hybrid Systems, Neuro-Genetic Hybrid Systems, Fuzzy-Genetic Hybrid Systems. **(10 Hrs.)**

Textbooks

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Principles of Soft Computing	S.N.Sivanandam and S.N.Deepa	Wiley	3rd (2018)
2	Fuzzy Logic with Engineering Applications	Timothy J. Ross	Wiley	3rd (2011)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Soft Computing & Intelligent Systems: Theory & Applications	N. K. Sinha, M. M. Gupta	Academic Press /Elsevier	1st (2009)
2	Fuzzy Logic and Genetic Algorithms: Synthesis & Applications	S.Rajasekaran, G. A. Vijayalakshmi	PHI	1st (2013)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-702(A)	Object-Oriented Analysis and Design	PEC	3	3	0	0	50	100	150

Course Outcomes:

At the end of the course the students will be able to:	
CO1	Explain the concepts of OOAD, UML and SDLC in software development.
CO2	Apply use case modeling for different software development scenarios.
CO3	Apply OOP principles, design patterns, and UML for development.
CO4	Design optimized software systems with diverse architectural styles and patterns.
CO5	Analyze design models and apply OOP in coding and testing.

Detailed Syllabus

Section-A

UNIT 1: Introduction to Object-Oriented Analysis and Design (OOAD): Overview of object-oriented concepts and principles, Benefits of using object-oriented analysis and design, Introduction to the Unified Modeling Language (UML), Software development life cycle (SDLC) and OOAD. (10 Hrs.)

UNIT 2: Requirements Gathering and Analysis: Understanding and documenting user requirements, use-case modeling and analysis, Domain modeling and identifying objects, classes, and relationships, Creating a requirements specification document. **(10 Hrs.)**

UNIT 3: Object-Oriented Design: Class and object modeling using UML, Object-oriented principles: inheritance, polymorphism, encapsulation, and abstraction, Design patterns and their application, Component-based design and modularization. **(12 Hrs.)**

Section-B

UNIT 4: System Architecture and Design: Architectural styles and patterns, designing software systems with multiple components and layers, User interface design and usability considerations, Designing for performance, scalability, and maintainability. **(10 Hrs.)**

UNIT 5: Implementation and Testing: Translating design models into code, Object-oriented programming languages and frameworks Testing strategies and techniques, Integration testing and system testing.

(10 Hrs.)

Textbooks

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Object-Oriented Analysis and Design with Application	Ph.D. Young, Bobbi , Grady Booch , Jim Conallen , Kelli Houston, Michael Engle, Robert Maksimchuk, John Fuller	Addition Wesley	3rd (2017)
2	Object-Oriented Analysis and Design with UML	Grady Booch, Robert A. Maksimchuk, Michael W. Engel, Bobbi J. Young, Jim Conallen, Kelli A. Houston	Pearson Education	3rd (2007)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Object-Oriented Analysis and Design	Brahma, Ramnath	Springer	2nd (2017)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-702 (B)	Software Testing and Automation	PEC	3	3	0	0	50	100	150

Course Outcomes:

At the end of the course the students will be able to: -	
CO1	Explain software testing principles, processes, and defect classification in SDLC.
CO2	Analyze manual test case design techniques and conceptualize user acceptance testing.
CO3	Describe test automation concepts, frameworks, and build knowledge of automation tools.
CO4	Design proficient test automation solutions with popular tools and integration.
CO5	Evaluate advanced testing methods for assessing performance and security for different applications.

Detailed Syllabus

Section-A

Unit 1: Introduction to Software Testing: Overview of software testing principles and concepts, Testing process and its phases, Software development life cycle and testing, Types of software defects and their classification. **(8 Hrs.)**

Unit 2: Manual Testing Techniques: Test case design and test coverage, Black-box testing techniques (equivalence partitioning, boundary value analysis, etc.), White-box testing techniques (statement coverage, branch coverage, etc.), User acceptance testing and usability testing. **(8 Hrs.)**

Unit 3: Test Automation Fundamentals: Introduction to test automation and its benefits, Automation framework types (linear, modular, data-driven, keyword-driven, etc.), Selecting appropriate test automation tools, Test automation scripting and execution. **(10 Hrs.)**

Section-B

Unit 4: Test Automation Tools and Frameworks: Introduction to popular test automation tools (e.g., Selenium, Appium), Creating and executing automated test scripts, Test data management and automation, Continuous integration, and test automation. (11 Hrs.)

Unit 5: Advanced Testing Topics: Performance testing and load testing, security testing and vulnerability assessment, mobile application testing, test management and defect tracking. **(8 Hrs.)**

Textbooks

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Software Testing: Principles and Practices	Srinivasan Desikan	Pearson Education	1st (2005)
2	Software Testing: Concepts and Operations	Ali Mili, Fairouz Tchier	Wiley	3rd (2015)

Reference Books

S.No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Fundamentals of Software Testing	ISTQB Certification	Cambridge	2nd (2017)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-702(C)	Applied Data Science	PEC	3	3	0	0	50	100	150

Course Outcomes:

At the end of the course the students will be able to:	
CO1	Describe the importance of data science and the various operations involved.
CO2	Articulate the different steps involved in the data acquisition ecosystem and their relevance.
CO3	Assess diverse machine learning techniques for data analysis for different applications.
CO4	Develop data visualization and analytics skills using different libraries and tools.
CO5	Apply various AI models and techniques for Text and Natural Language Processing

Detailed Syllabus

Section A

Unit 1: Introduction to Data Science: Overview of data science and its applications, Introduction to data analysis using Python and Jupyter notebooks, Data preprocessing and cleaning techniques, Exploratory data analysis and visualization. (**9 Hrs.)**

Unit 2: Data Wrangling and Preparation: Data acquisition and integration from various sources, Handling missing data and outliers, Feature engineering and transformation, Data quality assessment and validation. **(10 Hrs.)**

Unit 3: Machine Learning Techniques: Introduction to supervised and unsupervised learning, Regression and classification algorithms, Clustering and dimensionality reduction, Model evaluation and selection. (**11Hrs.)**

Section B

Unit 4: Data Visualization and Communication: Data visualization principles and best practices, Visualizing data using Python libraries (e.g., Matplotlib, Seaborn), Storytelling with data, Communicating data insights effectively. (**10 Hrs.)**

Unit 5: From Text to Deep Learning: Text mining and natural language processing, Time series analysis and forecasting, Recommendation systems, Deep learning, and neural networks. (**11 Hrs.)**

Textbooks

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Python for Data Analysis	Wes McKinney	O'Reilly Media	2nd (2017)
2	Data Analytics using Python	Bharti Motwania	Wiley	1st (2020)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Data Science and Machine Learning using Python	Dr. Reema Thareja	McGraw Hills	(2017)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-703	Professional Elective 3* (SWAYAM/NPTEL MOOC/ Any Other Platform)	MCC*	3	0	0	0	-	100	100

Course Outcomes: Defined by the online platform for the specific course chosen.

The department shall offer the SWAYAM / NPTEL course/Any other platform (12 weeks) out of the list of courses offered by these platforms around the time of commencement of the semester. However, the selected course should not be similar to the regular courses offered as a part of the department curriculum. The overall monitoring of the course will be under the supervision of the faculty in charge of the department.

The marks obtained by the student in the certification course will be tabulated by the concerned department.

Note: - In case the student does not pass the certification, exam or remains absent in the proctored examination, no certificate will be given to the candidate by the NPTEL/SWAYAM/Any other platform and the student will be deemed to have failed in the course. The student must appear again in the NPTEL/SWAYAM/Any other platform examination conducted for the same course or any other course as per the next semester schedule of NPTEL/SWAYAM/Any other platform and pass the examination.

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
OEC-703(A)	Strategic Management	OEC	3	3	0	0	50	100	150

COURSE OUTCOMES

At the end of the course the student will be able to:	
CO1	Explain various perspectives and concepts in the field of Strategic Management.
CO2	Analyze the competitive situation and strategic dilemma in dealing with dynamic global business environments.
CO3	Develop knowledge in strategy formulation and implementation in organizations
CO4	Articulate the knowledge of strategic evaluation and control**.**
CO5	Apply strategic management concepts in solving case studies

Detailed Syllabus

Section A

UNIT I: **Concept of Strategy**: Strategic Decision Making, Strategic Management Process. Difference between Policy, Strategy and Tactics. Strategic Intent, Vision, Mission, Goals. and Objectives. Environmental Scanning: Factors, Approaches, Methods, Environmental Appraisal. Internal Analysis / Organizational Appraisal. **(9 Hrs.)**

UNIT II: Corporate Level Strategies: Expansion, Stability, Retrenchment, Combination Business Level Strategies; Strategic Analysis and Choice; Corporate Portfolio Analysis- SWOT Analysis, BCG Matrix, GE Nine Cell Matrix. **(10 Hrs.)**

UNIT III: S**trategy Implementation**: Procedural implementation, Structural implementation, Structure and strategy Organizational Design and Change, Behavioral implementation, Corporate Politics and Power in Strategic Management, Corporate Ethics and Personal Values for Strategic Management. **(11 Hrs.)**

Section B

UNIT IV: Strategic Evaluation and Control: Strategy Review, Strategic Evaluation and Control; Contemporary Issues in Strategic Management: Balanced Scorecard and Blue Ocean Strategy. **(10 Hrs.)**

UNIT V: Case Study Analysis: Applying Strategic Management through the Case Method on the topic related to Strategic Management for Sustainability, Environmental Scanning and Organizational Scanning, Corporate and Business level Strategies and Contemporary issues in Strategic Management. **(11 Hrs.)**

Textbooks

S.No	Name of the Books	Name of Author	Publisher Name	Edition (Pub. Yr.)
1.	Strategic Management and Business Policy	Azhar Kazmi	Tata McGraw Hill	4th (2012)
2.	Strategic Management	Charles WL Hill and Gareth R Jones	Cengage Learning	11th (2012)

Reference Books

S.No	Name of the Books	Name of Author	Publisher Name	Edition& Pub Yr.
1	Strategic Management	Hill, Ireland, Hoskisson, Manikutty	Cengage Learning	9th (2016)
2	Strategic Management	Fred R David	PHI Learning	13th (2008)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
OEC-703(B)	Marketing Management	OEC	3	3	0	0	50	100	150

COURSE OUTCOMES

At the end of the course the student will be able to:

CO1	Describe the core concepts of marketing.
CO2	Articulate the value creation process focusing on the concepts of segmentation, targeting and positioning.
CO3	Comprehend the concepts of product and price Mix.
CO4	Examine the importance of the channel distribution and promotion decisions.
CO5	Interpret the consumer behavior model and the recent trends in marketing.

Detailed Syllabus

Section A

UNIT-I: Fundamentals of Marketing: Marketing – Concept, Importance, Scope; Concept of needs, wants and demands; Key Markets; Orientations of Marketing; Shifts in Marketing. Marketing Environment; Market Research Process, Marketing Plan, Marketing Productivity.
(9 Hrs.)

UNIT-II: Identifying Customers and Connecting with Customers: Marketing and Customer Value, Value Chain, Value delivery Process, Building Customer Value, Satisfaction and Loyalty, Customer Lifetime Value, Customer Relationship Management – Concept and Components. Identifying Market segments and Targets, Levels of Market Segmentation, Segmentation of Consumer Markets, Bases of Segmentation, Market Targeting & Positioning.
(10 Hrs.)

UNIT-III: Product and Price Decision: Product Mix – Concept, Product line and Product Mix. Branding, Packaging and labeling. Product life cycle. New product development process. Price Mix – Concept of Price and Pricing, Significance of Pricing Decision, Factors affecting Price Determination, Pricing Methods and Techniques, Pricing Policies and Strategies.
(11 Hrs.)

Section B

UNIT-IV: Place and Promotion Decision: Distribution Mix – Nature, Functions, and Types of Distribution Channels. Distribution Channel Intermediaries. Channel Management Decisions and Channel design. Promotion Mix – Concept and Role of Promotion, Determining Promotion Mix. Advertising, Personal selling, Sales Promotion, Publicity and Public Relations, Sales Promotion. **(10 Hrs.)**

UNIT-V: Analyzing the Demand and Trends in Marketing: Consumer Behavior – Concept and Factors Affecting. Buying Decision Process, Theories of Consumer Decision Making, Participants in Buying Process, Stages in Buying Process. Trends in Marketing – Digital Marketing, Mobile Marketing, Green Marketing, Social Media Marketing, Influencer Marketing, Affiliate Marketing, Omnichannel Marketing.
(10 Hrs.)

Textbooks

S. No	Name of the Suggested Books	Name of Author	Publisher Name	Edition(Pub. Yr.)
1	Marketing Management	Kotler, Keller, Koshy & Jha	Pearson Education	16th (2022)
2	Marketing management: Global perspective Indian context	Ramaswamy, V. S. & Namakumari, S	Macmillan, New Delhi	4th (2009)

Reference Books

S .No	Name of the Book	Name of Author	Publisher Name	Edition (Pub. Yr.)
1	MKTG - A South Asian Perspective	CW Lamb, J F Hair, Dheeraj Sharma, C McDaniel,	Cengage Learning, India	1st (2016)
2	Marketing Management	Rajan Saxena	Tata McGraw Hill	6th (2019)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
OEC-703(C)	Managerial Economics	OEC	3	3	0	0	50	100	150

COURSE OUTCOMES:

At the end of the course the student will be able to:	
CO1	Describe the demand concept as the basis of economic activity.
CO2	Articulate the various laws of production in both short and long run work for designing the optimum level output.
CO3	Analyze the various types of market structures.
CO4	Interpret the various pricing methods and strategies.
CO5	Evaluate the various macroeconomic concepts affecting the overall economy and undertake economic analysis.

Detailed Syllabus

Section A

Unit-I: Introduction to Managerial Economics: Circular Flow of Economic Activity, Demand Analysis and Estimation- Individual, Market and Firm Demand; Determinants of Demand; Demand Estimation; Business and Economic Forecasting; Elasticity of Demand. **(9 Hrs.)**

Unit-II: Law of Variable Proportions: Theory of Firms, Production Theory and Analysis- The Production Function; Economies of scale and Scope; Estimating the Production Function; Cost Theory and Analysis- The Economic Concept; Production and Cost; Short-run and long-run Cost Functions; Estimating Cost Functions. **(10 Hrs.)**

Unit-III: **Market Structure**: Price and Output determination under different market structures-Perfect Competition and Monopoly; Relevance of Perfect Competition and Monopoly; Monopolistic Competition, Oligopoly- Sweezy Model. **(9 Hrs.)**

Section B

Unit- lV: Pricing of Goods and Services: Pricing Factors, Pricing of Multiple Product, Product Bundling; Peak-load Pricing; Cost-plus or Mark-up Pricing; Cyclical Pricing; Price Discrimination; Degrees of Price Discrimination **(10 Hrs.)**

Unit-V: **Introduction to National Income**: National Income Concept, Models of National Income Determination; Economic Indicators; Business Cycles; Management of Cyclical Fluctuations, Monetary and Fiscal Policies; Economic Indicators; National Income; Inflation and Deflation. **(11 Hrs.)**

Textbooks

S.No	Name of the Books	Name of the Author	Publisher Name	Edition (Pub. Yr.)
1.	Managerial Economics: Economic tool for today decision makers	Paul G Keat Philip Young, Stephen Erfle	Pearson, Prentice Hall	7th (2013)
2.	Managerial Economics and Business Strategy	Michael Baye, Jeffry T. Prince	McGraw Hill	10th (2022)

Reference Books

S. No	Name of the Book	Name of the Author	Publisher Name	Edition (Pub. Yr.)
1.	Managerial Economics	Craig H.Peterson,W, Cris Lewis, Sudhir k Jain	Pearson Education	4th (2005)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-705	Seminar	PEC	2	0	0	4	-	100	100

Course Outcomes:

At the end of the course the students will be able to:	
CO1	Identify a relevant topic for the seminar from an emerging domain from the branch of study.
CO2	Conduct deep literature review from diverse sources including journals, whitepapers, credible web sources, industry reports, patent databases, books etc.
CO3	Put together material from diverse sources in the form of an engaging presentation with new insights.
CO4	Prepare a short whitepaper on the seminar topic providing a broad overview of the domain.
CO5	Make a high-quality presentation on the seminar topic and demonstrate competence in addressing queries from experts and the audience.

Guidelines and evaluation of Seminar

Students shall individually prepare and submit a seminar report on a topic of current relevance related to the field of Computer Science and Engineering. The reference shall include standard journals, conference proceedings, reputed magazines and textbooks, technical reports and URLs. Each student shall be evaluated by a team of at least three faculty members constituted by the Head of the Department based on presentation, technical content, adequacy of references, depth of knowledge and overall quality of the seminar report.

Distribution of Marks:

S. No	Evaluation Component	Marks
1	Seminar Report	40
2	Presentation	40
3	Attendance	20
Total Marks		100

Course Code	Course Name	Course Type	C d	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-711	Minor Project	PROJ	3	0	0	6	100	150	250

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Identify applied or research projects in consultation with the designated faculty/industry mentor addressing a relevant problem.
CO2	Critically examine work done in the domain to identify gaps if any.
CO3	Carry out detailed design of the project to help attain the final outcomes.
CO4	Implement the project as per the formulated design using modern tools and techniques.
CO5	Prepare a formal report and demonstrate deep understanding of all aspects of the project.

The project will be assigned to the students towards the end of 6th semester and will start working on these projects at the commencement of their 7th semester. The topic of the project will be decided as per the developments taking place in the field of Computer Science and Engineering.

This may require a complete literature survey, design, fabrication, simulation of models and/or some preliminary laboratory experiments etc. The same project can be extended to the 8th semester also.

Distribution of Marks:

S.No	Evaluation Component	Marks
1	Demonstration of Project	100
2	Viva-Voce	100
3	Project Report	50
	Total Marks	250

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
NCC-701	Essence of Indian knowledge and Tradition	NCC	0	0	0	2	Satisfactory/ Not Satisfactory		S/NS*

Course Outcomes:

At the end of the course the student will be able to:	
CO1	Explain the Vedic theism and Upanishadic concepts of Atman & Brahman
CO2	Acquire thorough knowledge about Carvaka, Jainism
CO3	Acquire thorough knowledge about Buddhism and comprehend the yoga philosophy
CO4	Comprehend epistemology of Nyaya-Vaisesika
CO5	Analyze Sankhya's Theory of Evolution

Detailed Syllabus

Section-A

Unit 1: Introduction to Indian Philosophy, Basic concepts of Vedic and the Upanishadic World Views: ṛta (The Cosmic Order); ṛṇa (Duty/Obligation) ātman; jāgrat, svapna, suṣupti, turīya; śreyas, preyas, karma**.** **(8 Hrs.)**

Unit 2: Cārvāka -Carvaka School: It's epistemology, Jainism: Concepts of sat, dravya, guṇa, paryāya, jīva, and ajīvaAnekāntavāda, syādvāda and saptabhangīnaya. **(6 Hrs.)**

Unit 3: Buddhism-Four Noble Truths, pratītyasamutpāda, kṣaṇabhangavāda, nairātmyavāda Yoga Philosophy: Organization of the YogaSutras. 2. Psychology of Yoga. Kinds of Klesas. 3. Eight-Fold Yoga. **(10 Hrs.)**

Section-B

Unit 4: Nyāya – Vaisesika -Nyāya -Epistemology: Four pramāṇa-s: Perception, Inference, Comparison and Testimony; Nature of Cause, Classification of Causes: samavāyi, asamavāyi, nimitta, asatkāryavāda? The Idea of God and proofs for His Existence. **(8 Hrs.)**

Unit 5: Sāṁkhya Duality: prakṛiti and its constituents, proofs for the existence of prakriti, puruṣa: It's nature and arguments for its existence, plurality of puruṣa. **(6 Hrs.)**

Textbooks

S. No	Name of the Suggested Books	Name of Author	Publisher Name	Edition (Pub. Yr.)
1	An Introduction to Indian Philosophy	S. C. Chatterjee & D. M. Dutta	Calcutta University Press, Kolkata	7th (1968)
2	Outline of Indian Philosophy	M. Hiriyanna	Motilal BanarsidassPubl New Delhi	5th (2014)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	A Critical Survey of Indian Philosophy	Chandradhar Sharma	Motilal BanarsidassPubl New Delhi	14th (2016)
2	Indian Philosophy	Sarvepalli Radhakrishnan	Oxford University press	2nd (2008)

Scheme-2

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Mid Semester	Final	Total
1	PII-701	Industry Internship-I	Professional Industry Internship	15	0	0	30	350	500	850

The internship will be assigned to the students towards the end of the 6th semester, and they will start working on those projects at the commencement of the 7th semester. The students shall submit the details of the company/Industry where they intend to do their project work along with the company consent letter in the 7th semester. The departmental academic Committee will finalize and approve the project and provide the allotment of internal guide to each project who will periodically monitor and evaluate the student performance during the project.

The topic of the projects will be decided as per the development taking place in the field of Computer Science and Engineering. This may require a complete literature survey along with design, fabrication, simulation of the models and/or preliminary laboratory experiments etc. The students will have to submit a detailed project report individually to their internal guide and a copy of the certificate, if awarded, should also be appended to the report. They should also submit a progress report of their project duly signed by the concerned Authority via mail to the respective guide.

Note: Students are also allowed to start their start up provided they submit a DPR with a detailed proposal of their startup that would define the action plan and idea to the start-up cell. Only after the submitted proposal has been approved by the start-up cell the students will be allowed to work on their project.

Guidelines for Evaluation of the Industry Internship in 7th semester:

The evaluation shall be carried out once in mid-semester and once at the end of the semester. The distribution of marks is given below:

Sr. No.	Distribution	Mid Semester	Final
1	Viva-Voce	150	150
2	Presentation	200	200
3	Report	-	150
Marks		350	500
Total Marks		**850**	

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
2	COM-703	Professional Elective 3* (SWAYAM/NPTEL MOOC/Any Other Platform)	MCC*	3	0	0	0	-	100	100

Course Outcomes: Defined by the online platform for the specific course chosen.

The department shall offer the SWAYAM / NPTEL course/Any other platform (12 weeks) out of the list of courses offered by these platforms around the time of commencement of the semester. However, the selected course should not be similar to the regular courses offered as a part of the department curriculum. The overall monitoring of the course will be under the supervision of the faculty in charge of the department.

The marks obtained by the student in the certification course will be tabulated by the concerned department.

Note: - In case the student does not pass the certification, exam or remains absent in the proctored examination, no certificate will be given to the candidate by the NPTEL/SWAYAM/Any other platform and the student will be deemed to have failed in the course. The student must appear again in the NPTEL/SWAYAM/Any other platform examination conducted for the same course or any other course as per the next semester schedule of NPTEL/SWAYAM/Any other platform and pass the examination.

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
3	COM-703	Professional Elective 4* (SWAYAM/NPTEL MOOC/ Any Other Platform)	MCC*	3	0	0	0	-	100	100

Course Outcomes: Defined by the online platform for the specific course chosen.

The department shall offer the SWAYAM / NPTEL course/Any other platform (12 weeks) out of the list of courses offered by these platforms around the time of commencement of the semester. However, the selected course should not be similar to the regular courses offered as a part of the department curriculum. The overall monitoring of the course will be under the supervision of the faculty in charge of the department.

The marks obtained by the student in the certification course will be tabulated by the concerned department.

Note: - In case the student does not pass the certification, exam or remains absent in the proctored examination, no certificate will be given to the candidate by the NPTEL/SWAYAM/Any other platform and the student will be deemed to have failed in the course. The student must appear again in the NPTEL/SWAYAM/Any other platform examination conducted for the same course or any other course as per the next semester schedule of NPTEL/SWAYAM/Any other platform and pass the examination.

SCHEME –I
SEMESTER 8

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-801(A)	Advanced Machine learning	PEC	3	3	0	0	50	100	150

Course Outcomes:

At the end of the course the students will be able to:	
CO1	Apply probabilistic learning techniques for optimizing AI models
CO2	Articulate the application of decision trees for solving classification and regression problems.
CO3	Develop understanding of optimization techniques for ANNs.
CO4	Apply Deep Learning Architectures for a specific class of applications.
CO5	Appreciate reinforcement learning technique and its applications to specific use-cases.

Detailed Syllabus

Section-A

Unit 1: Probabilistic Model: Probability theory: sample space, probability function, conditional probability, Bayesian Learning Probability theory and Bayes rule, Naive Bayes learning algorithm, Bayes nets. **(9 Hrs.)**

Unit 2: Representation of Decision Tree: Decision Trees concepts, Recursive induction of decision trees, best splitting attribute: entropy and information gain. Searching for simple trees and computational complexity, Overfitting and underfitting, Noisy data, and Data pruning. Random Forest approach. **(10 Hrs.)**

Unit 3: Artificial Neural Network (ANN): Introduction to ANN, Perceptron, Cost Function, Gradient Checking, Multi-Layer Perceptron and Backpropagation Algorithm, Neural Network, Random Initialization. **(10 Hrs.)**

Section B

Unit 4: Introduction to Deep Learning Architectures: Overview of deep learning architectures and their advantage, Understanding the depth, width, and capacity of neural networks. Concept of autoencoders for unsupervised learning and dimensionality reduction. **(10 Hrs.)**

Unit 5: Reinforcement Learning: Introduction to Reinforcement Learning (RL), Markov Decision Process (MDP) formulation, Components of RL: Agent, Environment, State, Action, Reward, Policy, Value function, Function approximation. **(9 Hrs.)**

Textbooks

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Machine Learning	Tom Mitchell	McGraw Hill	1st (1997)
2	Machine Learning: a Probabilistic Perspective	Kevin P. Murphy	The MIT Press	1st (2012)

Reference Book

S.No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Introduction to Machine Learning	Jeeva Jose	Khanna Book Publishing	1st (2020)
2	Introduction to Machine Learning,	Ethem Apaydin,	The MIT Press	2nd (2010)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-801(B)	Generative Artificial Intelligence (AI)	PEC	3	3	0	0	50	100	150

Course Outcomes:

At the end of the course the student will be able to: -	
CO1	Explain the foundational concepts of Generative Artificial Intelligence.
CO2	Analyze various AI based generative models and their applications.
CO3	Describe GANs and their applicability to various unsupervised learning problems.
CO4	Evaluate RNNs and LSTM models for sequential processing in Natural Language Processing.
CO5	Explore use of pre-trained language models including metrics for evaluating generative models.

Detailed Syllabus

Section-A

Unit 1: Introduction: Overview of Generative AI (GenAI) and its applications in various domains, Understanding the difference between generative and discriminative models, Taxonomy of generative models, GenAI modalities mappings. **(10 Hrs.)**

Unit 2: Generative Models: Fundamentals and Techniques, Introduction to basic generative models (Gaussian Mixture Models (GMMs) and Hidden Markov Models (HMMs)), Latent Variable Models, Statistical Language Models (SLM), Role of encoder and decoder networks. **(10 Hrs.)**

Unit 3: Generative Adversarial Networks (GANs): Introduction to GANs and the GAN framework, Generator-discriminator adversarial training process, Variants of GANs: Conditional GANs, Wasserstein GANs. Autoregressive Models (PixelCNN and PixelRNN architectures for image generation). **(10 Hrs.)**

Section-B

Unit 4: Sequence Generation and Transformers: Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) cells, Introduction to Transformers: Motivation for Transformers and their advantages over RNNs, Transformer architecture: self-attention and feed-forward layers, multi-head attention mechanism. (**10 Hrs.)**

Unit 5: Language Generation with Transformers: Using pre-trained language models like GPT (Generative Pre-trained Transformer), Conditional language generation, Metrics and techniques for evaluating generative models. **(10 Hrs.)**

Textbooks

S.No	Name of the Suggested Books	Name of Author	Publisher Name	Edition (Pub. Yr.)
1	Generative AI with python	Joseph Babcock, Raghav Bali	Packt	1st (2021)
2	ChatGPT for Thought Leaders and Content Creators: Unlocking the	Dr. Gleb Tsipursky	Intentional Insights	1st (2023)

	Potential of Generative AI for Innovative and Effective.			

Reference Books

S.No.	Name of the Book	Name of Author	Publisher Name	Edition (Pub. Yr.)
1	Exploring Deepfakes	Bryan Lyon (Author), Matt Tora	Packt Publishing	1st (2023)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-801(C)	Knowledge Representation and Reasoning	PEC	3	3	0	0	50	100	150

Course Outcomes

At the end of the course the students will be able to	
CO1	Describe the modern view of Artificial Intelligence (AI).
CO2	Comprehend the knowledge representation techniques in AI.
CO3	Apply the knowledge representations in real-world problem-solving.
CO4	Analyze the control knowledge to make informed decisions.
CO5	Develop AI-driven chatbots and virtual assistants.

Detailed Syllabus

Section-A

UNIT 1: Introduction: - The AI problems, AI techniques, The level of the model, Criteria for success, AI tasks. Problems, Problem Spaces & Research, Defining problem as a state space search, Production system, Problem characteristics, Production system characteristics, Issues in the design of search programs. **(9 Hrs.)**

UNIT 2: Knowledge Representation: - Representation and mappings, approaches to knowledge representation, Issues of knowledge representation, the frame problem, Semantic networks. **(9 Hrs.)**

UNIT 3: Representing Knowledge using Rules: -Procedural Vs. declarative knowledge, Logic programming, Forward Vs. backward searching, Matching, Control knowledge, Heuristic search techniques: - Generate & test, Hill climbing, Best First Search, Problem reduction, Constraint satisfaction, Means and analysis. **(11 Hrs.)**

Section B

UNIT 4: Logic: - Representing simple facts, Its Logic representing instances and its relationship, Computable Functions & Predicates, Resolution, Natural Deduction, Conversion to Clause Form. **(11 Hrs.)**

UNIT 5: Reasoning: - Introduction to non-monotonic reasoning, Logics for non- monotonic reasoning, Implementation Issues, Augmenting a Problem Solver, Implementation by a) Depth - First Search b) Breadth - First Search. Statistical Reasoning, Dempster Shafer Theory, Fuzzy logic, Introduction to expert system development. **(10 Hrs.)**

Textbooks

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Artificial Intelligence	Elaine Rich, Kevin Knight and Shivshankar B Nair	McGraw Hill Education	3rd (2017)
2	Artificial Intelligence: A modern approach	Stuart Russel	Pearson Education	3rd (2010)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Knowledge Representation and Reasoning	Ronald Brachman & Morgan Kaufmann	McGraw Hill Education	1st (2004)
2	Artificial Intelligence	Padhy	Oxford Press	1st (2005)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-802 (A)	UAVs and Drone Technology	PEC	3	2	1	0	50	100	50

Course Outcomes:

At the end of the course the students will be able to	
CO1	Articulate the basic principles, components, and applications of drone technology.
CO2	Demonstrate knowledge of drone hardware components and their functionality.
CO3	Appreciate drone flight dynamics, control systems, and navigation techniques.
CO4	Develop knowledge of drone navigation and synchronization techniques.
CO5	Formulate algorithms for drone operations, control, and intelligence.

Detailed Syllabus

Section-A

UNIT 1: Introduction to Drones: Basics of drone technology and its historical background, Types of drones and their applications in different industries, Current trends, and prospects of drone technology. Drone applications in industries such as agriculture, construction, cinematography. **(6 Hrs.)**

UNIT 2: Drone Components and Systems: Understanding drone hardware components: frame, motors, propellers, and batteries, Introduction to drone sensors like GPS, accelerometers, gyroscopes, and camera. **(9 Hrs.)**

UNIT 3: Overview of drone software and control: Drone Flight Dynamics and Control, Principles of flight including aerodynamics, lift, drag, and thrust, Drone stabilization and control mechanisms, Introduction to flight modes and maneuvers. **(11 Hrs.)**

Section-B

UNIT 4: Drone Navigation and Synchronization: Global Positioning System (GPS) and its role in drone navigation, Autonomous flight and obstacle avoidance systems, Sensor technologies for mapping, imaging, and data collection, Drone Operation and Flight Planning, Practical Implementation of application-based Drone Model. **(9 Hrs.)**

UNIT 5: Algorithms: Flight Control Algorithms, Navigation Algorithms, Path Planning Algorithms, Obstacle Detection and Avoidance, Target Tracking Algorithms, Swarm Algorithms, Image, and Video Processing. **(8 Hrs.)**

Textbooks

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Introduction to UAV Systems	Paul Fahlstrom and Thomas Gleason	Wiley Publication	4th (2012)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Drone Technology: Future Trends and Practical Applications	Sachi Nandan Mohanty	Wiley Publication	1st (2023)
2	Drone Technology in Architecture, Engineering and Construction	Daniel Tal, John Altschuld	Wiley Publication	1st (2021)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-802(B)	Smart City Technologies	PEC	3	3	0	0	50	100	150

Course Outcomes:

At the end of the course the students will be able to	
CO1	Build a deep understanding of the Smart City concept and the supporting policy framework.
CO2	Evaluate the use of Data Analytics and AI in diverse aspects of smart city management.
CO3	Describe smart transportation systems for smart cities; their design and operational framework.
CO4	Develop competence in applying the IoT Framework to Smart City operations.
CO5	Explain the importance of sustainability w.r.t waste management, green buildings, and water management systems.

Detailed Syllabus

Section-A

Unit 1: Introduction to the Smart Cities: Definition, characteristics and importance of Smart Cities, Overview of Urban Development and challenges in India, Role of Technology in transforming urban spaces, India 100 smart cities policy and mission, Government initiatives and policies for smart city development, Public-private partnerships for implementing smart city projects. **(9 Hrs.)**

Unit 2 Data Analytics and AI for Smart Cities: Big data analytics for urban planning and management, AI applications in smart cities for decision-making, Machine learning algorithms for urban data analysis. **(11 Hrs.)**

Unit 3: Smart Transportation and Mobility: Intelligent transportation systems (ITS) for congestion management, Smart public transportation and mobility solutions, Electric vehicles and sustainable transportation options. **(8 Hrs.)**

Section B

Unit 4: IoT and Sensor Networks: Introduction to IoT and its applications in smart cities, Smart sensors and data collection techniques and Integration of IoT in urban systems and services. **(9 Hrs.)**

Unit 5: Environmental Sustainability in Smart Cities: Waste Management and recycling strategies, green buildings and sustainable urban development, water management and conservation practices, Renewable energy sources and their integration into urban settings, Smart grids and energy management systems, Building automation and energy-efficient technologies. **(8 Hrs.)**

Textbooks

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Introduction to Smart Cities	Anil Kumar	Pearson	1st (2019)
2	Smart Cities: Foundations, Principles, and Applications	Houbing Song, Ravi Srinivasan, Tamim Sookoor, Sabina Jeschke	Wiley	1st (2017)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	The Smart City Transformations: The Revolution of the 21st Century	Amitabh Satyam & Igor Clazada	Bloomsbury Academic	1st (2020)

Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
							Sessional	Final Exam	Total
COM-802(C)	Foundations of Internet of things and Industry 4.0	PEC	3	3	0	0	50	100	150

COURSE OUTCOMES

At the end of the course the student will be able to: -	
CO1	Describe the importance of Industry 4.0 for meeting future societal needs.
CO2	Develop critical understanding on cyber-physical systems amalgamating sensors, AI, Big Data analytics and cybersecurity and their applications.
CO3	Explain the architecture of Industrial Internet-of-Things.
CO4	Evaluate the structure operations of Industrial Internet-of-Things through supporting technologies such as SDNs and Big Data Analytics.
CO5	Appraise the diverse application of Industry 4.0 and related management concepts.

Detailed Syllabus
Section-A

Unit 1: Introduction: Sensing & actuation, Industry 4.0: Globalization and Emerging Issues, The Fourth Revolution, LEAN Production Systems, Smart and Connected Business Perspective, Smart Factories. **(9 Hrs.)**

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Unit 2: Industry 4.0: Cyber Physical Systems and Next Generation Sensors, Collaborative Platform and Product Lifecycle Management, Augmented Reality and Virtual Reality, Artificial Intelligence, Big Data and Advanced Analysis. Cybersecurity in Industry 4.0, Basics of Industrial IoT: Industrial Processes, Industrial Sensing & Actuation, Industrial Internet Systems.

(9 Hrs.)

Unit 3: Industrial IoT: Business Model and Reference Architecture: IIoT-Business Models, IIoT Reference Architecture, Industrial IoT- Layers: IIoT Sensing, IIoT Processing, IIoT Communication, Industrial IoT- Layers: IIoT Communication, IIoT Networking. **(9 Hrs.)**

Section B

Unit 4: Industrial IoT: Big Data Analytics and Software Defined Networks: IIoT Analytics - Introduction, Machine Learning and Data Science, Data Management with Hadoop, Industrial IoT: Big Data Analytics, Software Defined Networks: SDN in IIoT, Data Center Networks, Industrial IoT: Security and Fog Computing: Cloud Computing, Security in IIoT, Industrial IoT- Application Domains: Factories and Assembly Line, Food Industry. **(9 Hrs.)**

Unit 5: Industrial IoT- Application Domains: Healthcare, Power Plants, Inventory Management & Quality Control, Plant Safety and Security (Including AR and VR safety applications), Facility Management, Oil, chemical and pharmaceutical industry, Applications of UAVs in Industries, Real case studies like Milk Processing and Packaging Industries, Manufacturing Industries. **(9 Hrs.)**

Textbooks

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Introduction to Industrial Internet of Things and Industry 4.0	S. Misra, C. Roy, and A. Mukherjee	CRC Press	1st (2020)
2	Industry 4.0: The Industrial Internet of Things	Alasdair Gilchrist	Apress	1st (2019)

Reference Books

S. No.	Name of the Books	Author	Publisher	Edition (Pub. Yr.)
1	Introduction to IoT	S. Misra, A. Mukherjee, and A. Roy	Cambridge University Press	1st (2020)

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
7	COM-803	Professional Elective 3 (SWAYAM/NPTEL/ Any other MOOC Platform)	MCC*	3	0	0	0	-	100	100

Course Outcomes: Defined by the online platform for the specific course chosen.

The department shall offer the SWAYAM / NPTEL course/Any other platform (12 weeks) out of the list of courses offered by these platforms around the time of commencement of the semester. However, the selected course should not be similar to the regular courses offered as a part of the department curriculum. The overall monitoring of the course will be under the supervision of the faculty in charge of the department.
The marks obtained by the student in the certification course will be tabulated by the concerned department.

Note: - In case the student does not pass the certification, exam or remains absent in the proctored examination, no certificate will be given to the candidate by the NPTEL/SWAYAM/Any other platform and the student will be deemed to have failed in the course. The student must appear again in the NPTEL/SWAYAM/Any other platform examination conducted for the same course or any other course as per the next semester schedule of NPTEL/SWAYAM/Any other platform and pass the examination.

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Sessional	Final Exam	Total
8	COM-811	Major Project	PROJ	6	0	0	12	200	300	500

Project Outcomes:

After completion of this project the student will be able to	
CO1	Identify and formulate the project outline addressing a relevant problem.
CO2	Perform detailed literature survey of the solutions pertaining to the identified problem and identify gaps.
CO3	Undertake the design and development of the project applying formal design and implementation techniques.
CO4	Prepare a formal report describing the project in sufficient technical detail as per prescribed format.
CO5	Demonstrate the project proficiently and display deep understanding of the concepts and technology used.

The project will be assigned to the students towards the end of the 7th semester, and they will start working on those projects at the commencement of their 8th semester. The student will submit the synopsis of their project work in the 7th semester. The departmental level committee will finalize and approve the projects. However, a departmental guide will be allocated to each project who shall periodically evaluate the student's performance during the project.

The topic of the project will be decided as per the developments taking place in the field of Computer Science and Engineering. This may require a complete literature survey, design, fabrication, simulation of some models and/or some preliminary laboratory experiments etc. The student will have to submit a detailed project report individually to the internal guide and a copy of the certificate should also be appended to the report:

Guidelines for Evaluation of the Major Project in 8th semester:

The evaluation shall be carried out once in mid-semester and once at the end of the semester. The distribution of marks is given below:

Sr. No.	Distribution	Mid Semester	Final
1	Project Demonstration	100	100
2	Project Design	100	NA
3	Implementation and Functionality	-	100
4	Documentation and Technical Writing	-	100
Marks		200	300
Total Marks		500	

Scheme-2

S. No.	Course Code	Course Name	Course Type	Cd	L	T	P	Marks		
								Mid Semester	Final	Total
1	PII-801	Industry Internship	Professional Industry Internship	15	0	0	30	300	600	900

Internship Outcomes:

After completion of the internship the student will be able to:	
CO1	Identify a relevant problem or functional area in consultation with industry expert or designated mentor and the expected outcomes.
CO2	Perform detailed analysis of the problem in line with expected outcomes.
CO3	Undertake the project work/field work to effectively attain the expected outcomes.
CO4	Demonstrate proficiency in professional communication, teamwork, and work ethics.
CO5	Prepare a formal report outlining the work done, outcomes and demonstrate deep understanding of the work done.

The internship will be assigned to the students towards the end of 7th semester, and they will start working on those projects at the commencement of the 8th semester. The students shall submit the details of the company/Industry where they intend to do their project work along with the company consent letter in the 8th semester. The departmental academic Committee will finalize and approve the project and provide the allotment of internal guide to each project who will periodically monitor and evaluate the student performance during the project.

The topic of the projects will be decided as per the development taking place in the field of the Computer Science and Engineering This may require a complete literature survey along with design, fabrication, simulation of the models and/or preliminary laboratory experiments etc.

The students will have to submit a detailed project report individually to their internal guide and a copy of the certificate, if awarded, should also be appended to the report. They should also submit a progress report of their project duly signed by the concerned Authority via mail to the respective guide.

Note: Students are also allowed to start their start up provided they submit a DPR with a detailed proposal of their startup that would define the action plan and idea to the start-up cell. Only after the submitted proposal has been approved by the start-up cell the students will be allowed to work on their project.

Guidelines for Evaluation of the Industry Internship in 8th semester:

The evaluation shall be carried out once in mid-semester and once at the end of the semester. The distribution of marks is given below:

Sr. No.	Distribution	Mid Semester	Final
1	Viva-Voce	150	150
2	Presentation	150	150
3	Report	-	300
Marks		300	600
Total Marks		900	